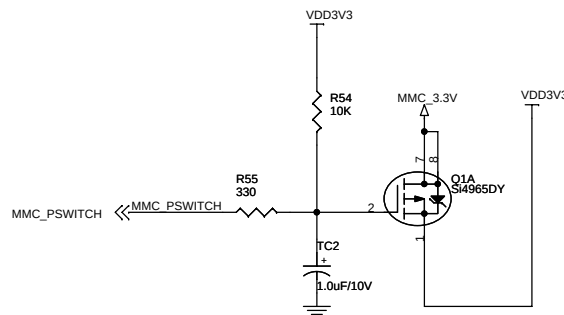
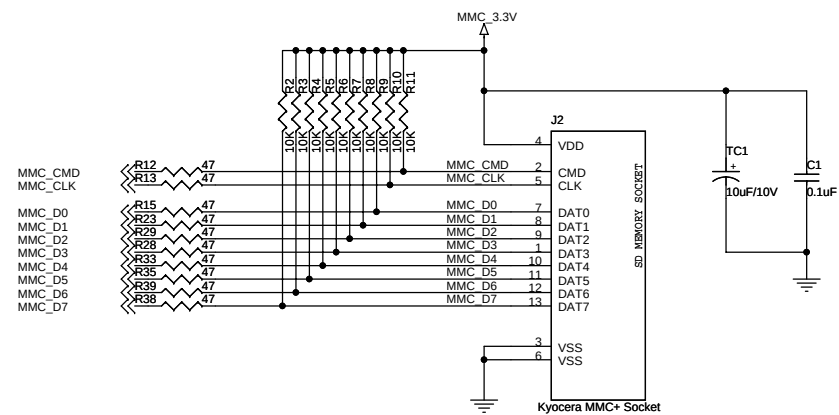
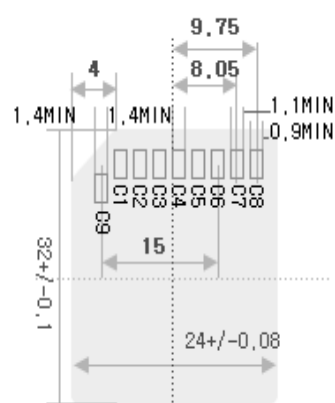
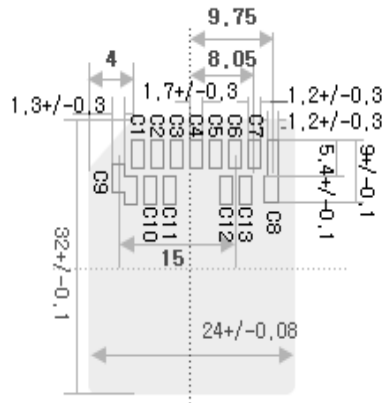




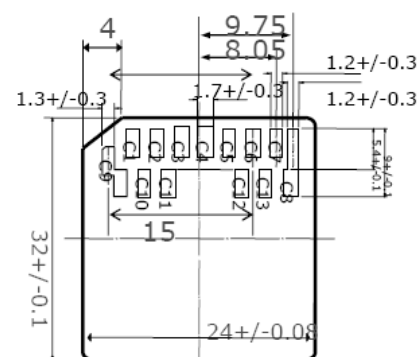
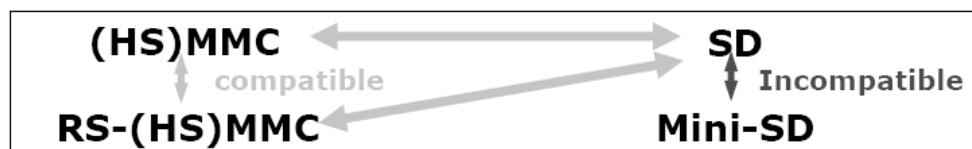
SD/HS-MMC SOCKET



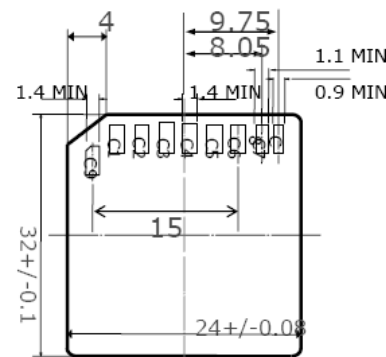
SD Card



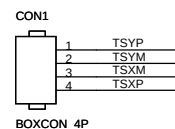
HS-MMC



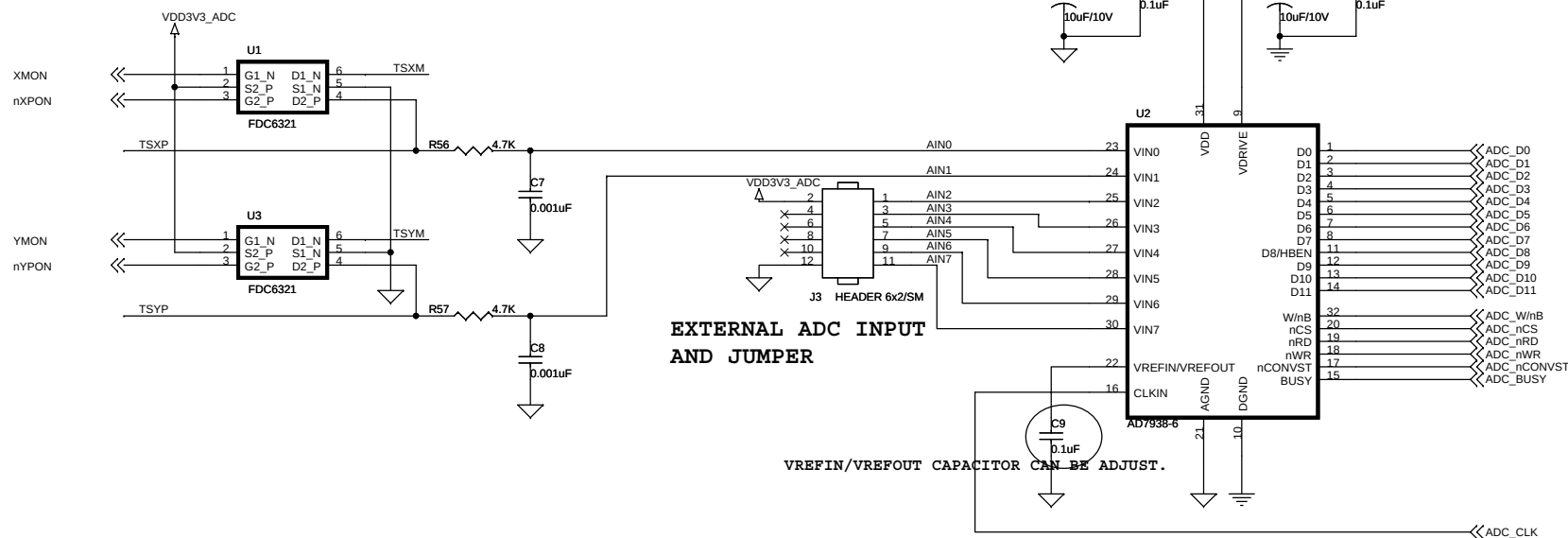
MMC



SD Card

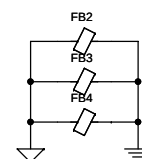
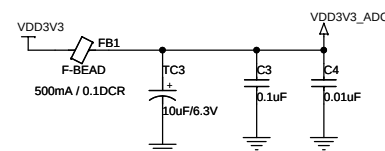


TOUCHSCREEN CONN.
MITSHMI
M60-04-30-134P



EXTERNAL ADC INPUT
AND JUMPER

VREFIN/VREFOUT CAPACITOR CAN BE ADJUST.



SANGHWA FPGA PLATFORM

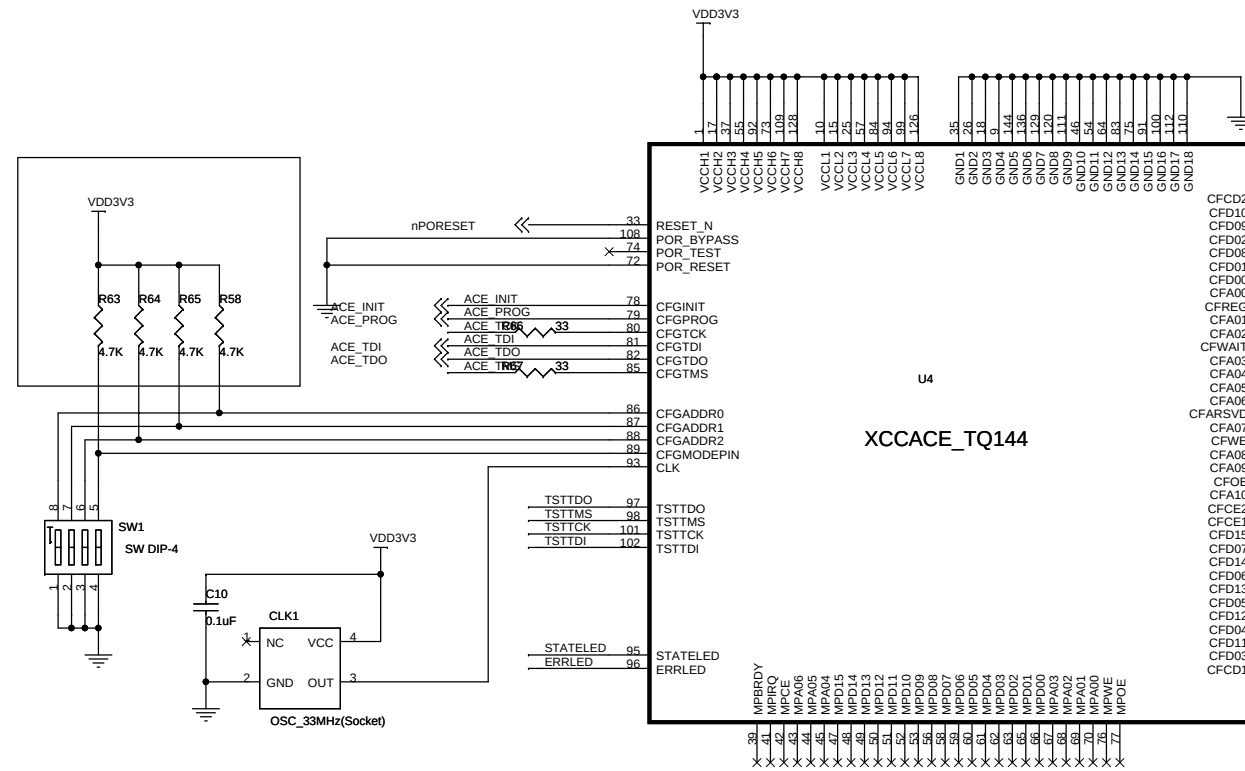
PAGE #	DESCRIPTION
P00	TITLE PAGE (THIS)
P01	SD/MMC/TOUCH
P02	SYSTEM ACE (FPGA CONFIGURATION)
P03	NONE
P04	VIDEO
P05	MEMORY BLOCK
P06	SW/AC97
P07	ETH/USB/UART
P08	CLOCK/PLL/MULTIPLEXOR
P09	SYSTEM POWER
P10	FPGA TEST MICTOR
P11	F0 CONFIG/CLOCK
P12	F0 BANK 1/2/5/6
P13	F0 BANK 7/8/9/10
P14	F0 BANK 11/12/13/14
P15	F0 BANK 15/16/GND
P16	F0 POWER VCCO
P17	F1 CONFIG/CLOCK
P18	F1 BANK 1/2/5/7
P19	F1 BANK 7/8/9/11
P20	F1 BANK 11/12/13/15
P21	F1 BANK 15/16/GND
P22	F1 POWER VCCO

2006-7-20
DESIGNED BY DDSMURF

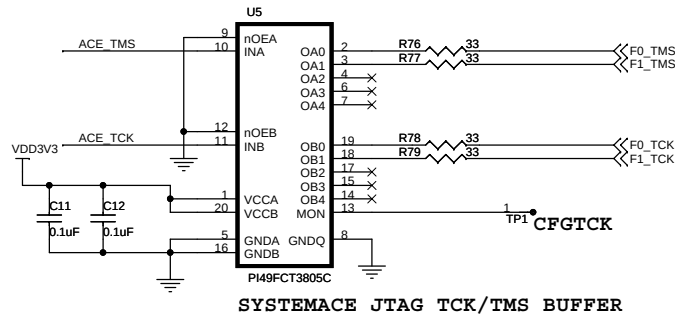
TitleSANGHWA FPGA PLATFORM SANGHWA FPGA PLATFORM		
Size A3	Document Number <Doc>	Rev 1.0
Date:	Saturday, July 22, 2006	Sheet 1 of 26

2006-3-3 부품 추가

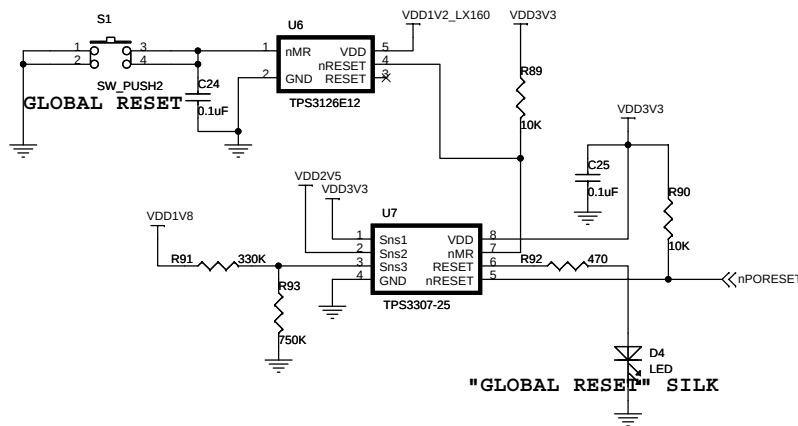
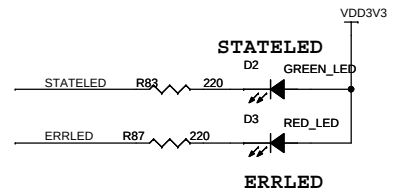
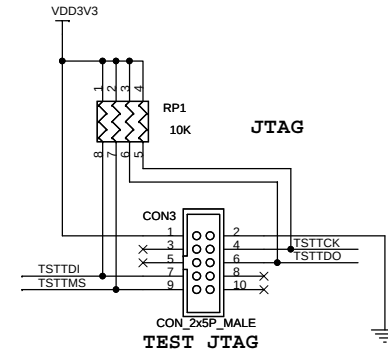
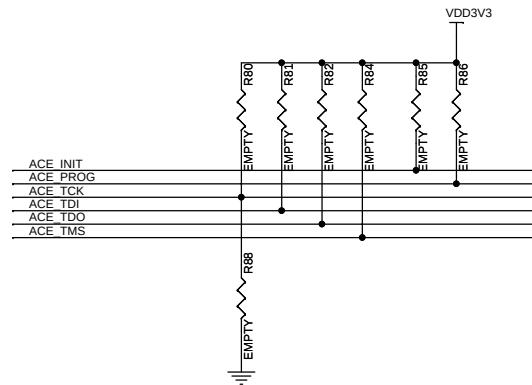
R184, R534, R535



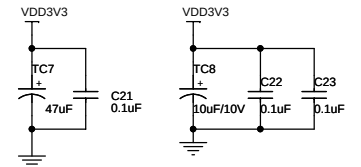
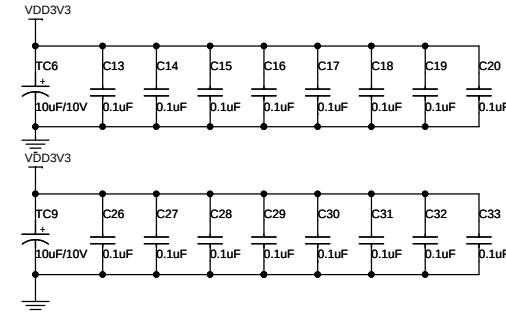
ACE Module

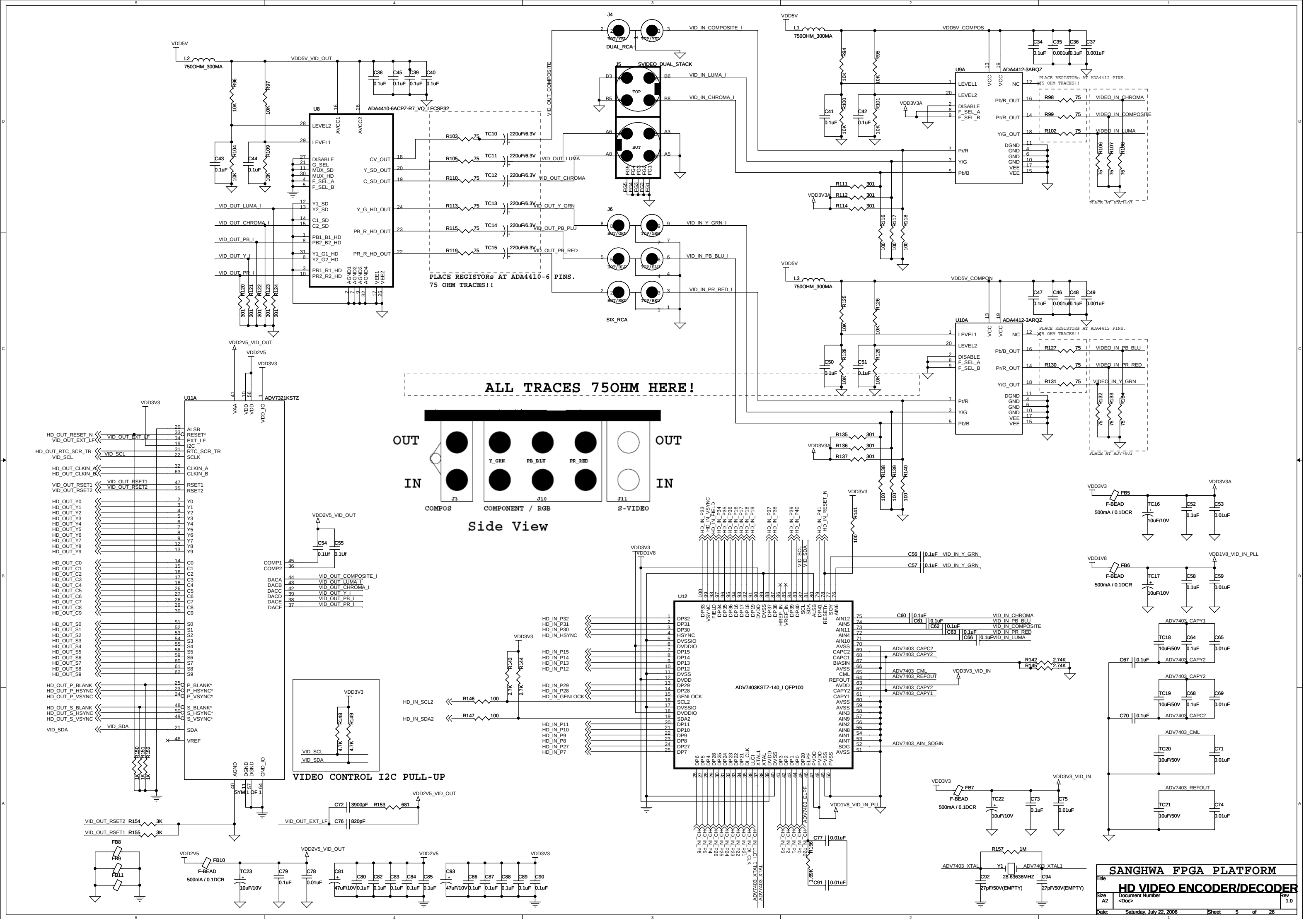


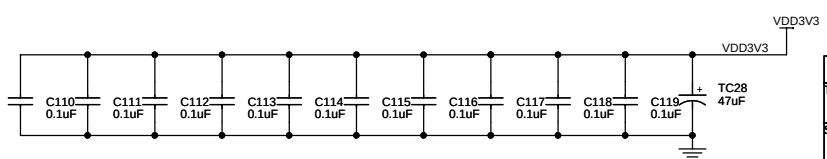
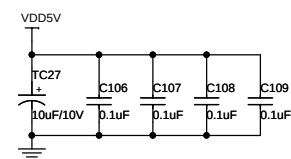
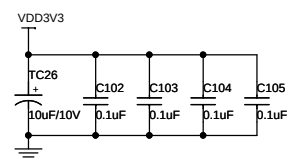
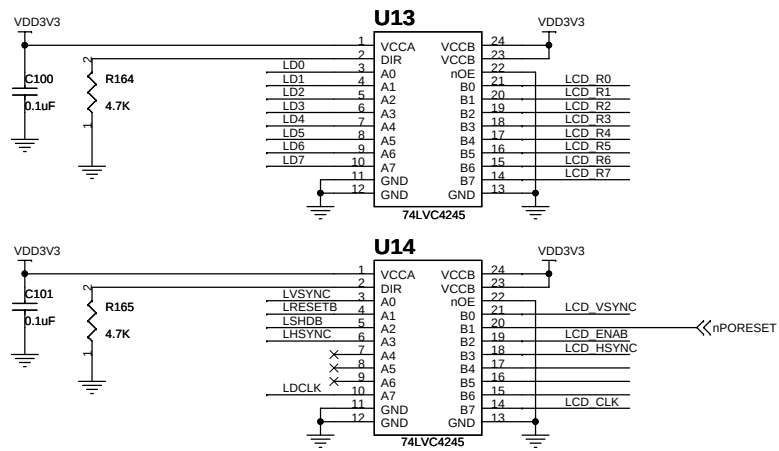
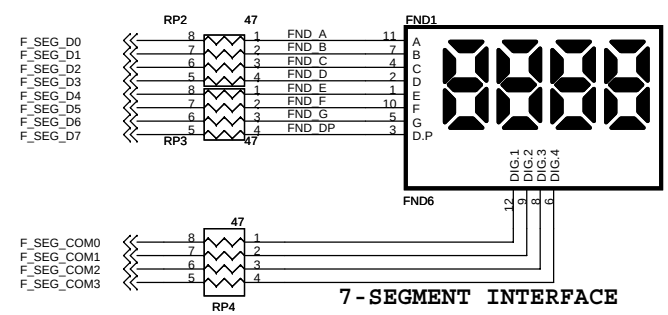
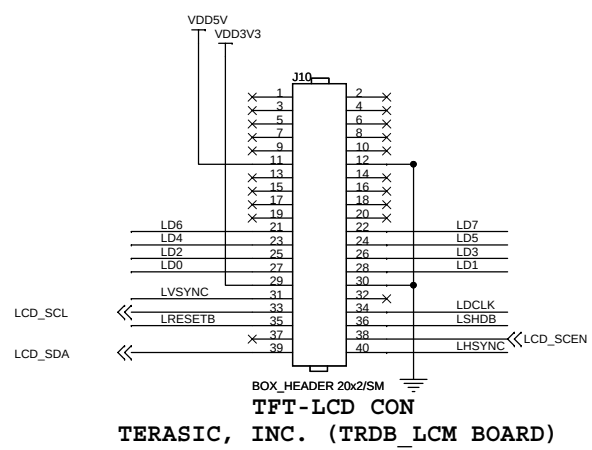
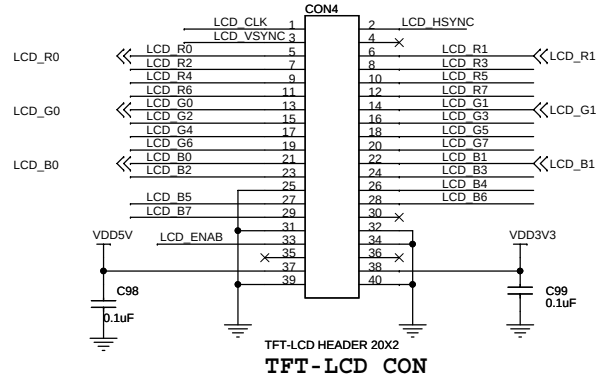
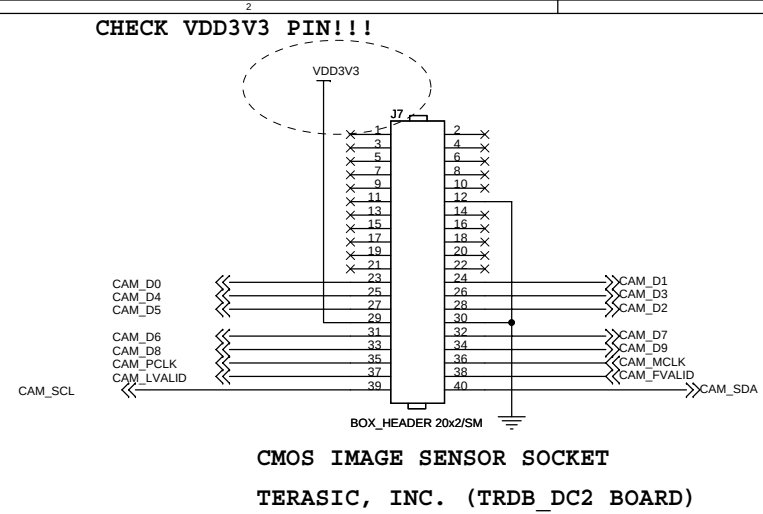
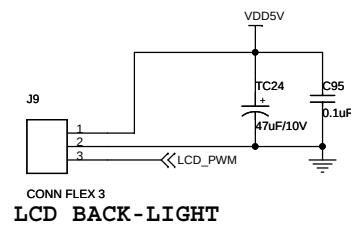
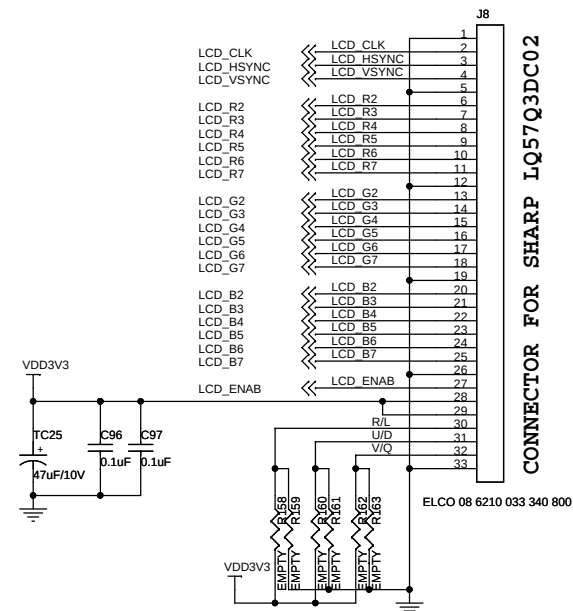
SYSTEMACE JTAG TCK/TMS BUFFER

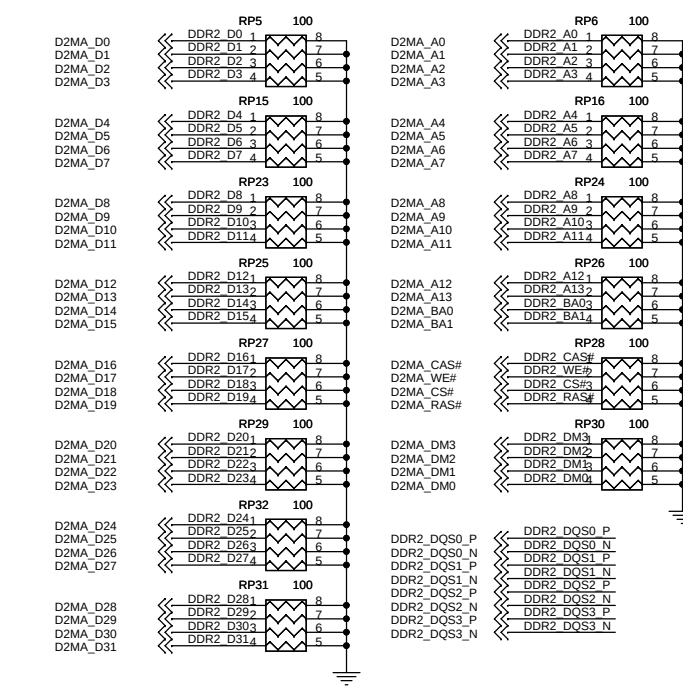


"GLOBAL RESET" SILK









PLACE CLOSE TO DDR MEMORY DEVICES

NOTE THAT CK & DQS HAVE DIFFERENTIAL TERMINATIONS.

DQ/DQS FOR COMPONENTS DO NOT HAVE ANY TERMINATION AT THE FPGA END.

THE INTENTION IS TO RELY ON SSTL18_II DCI I/O OF FPGA FOR TERMINATION.

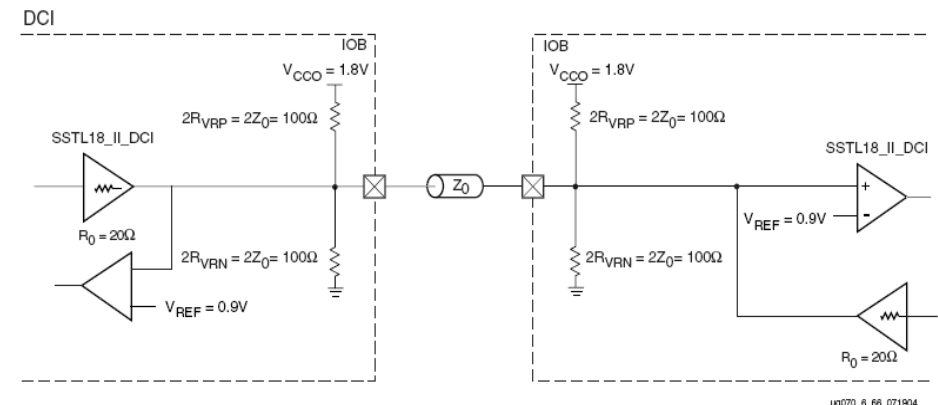
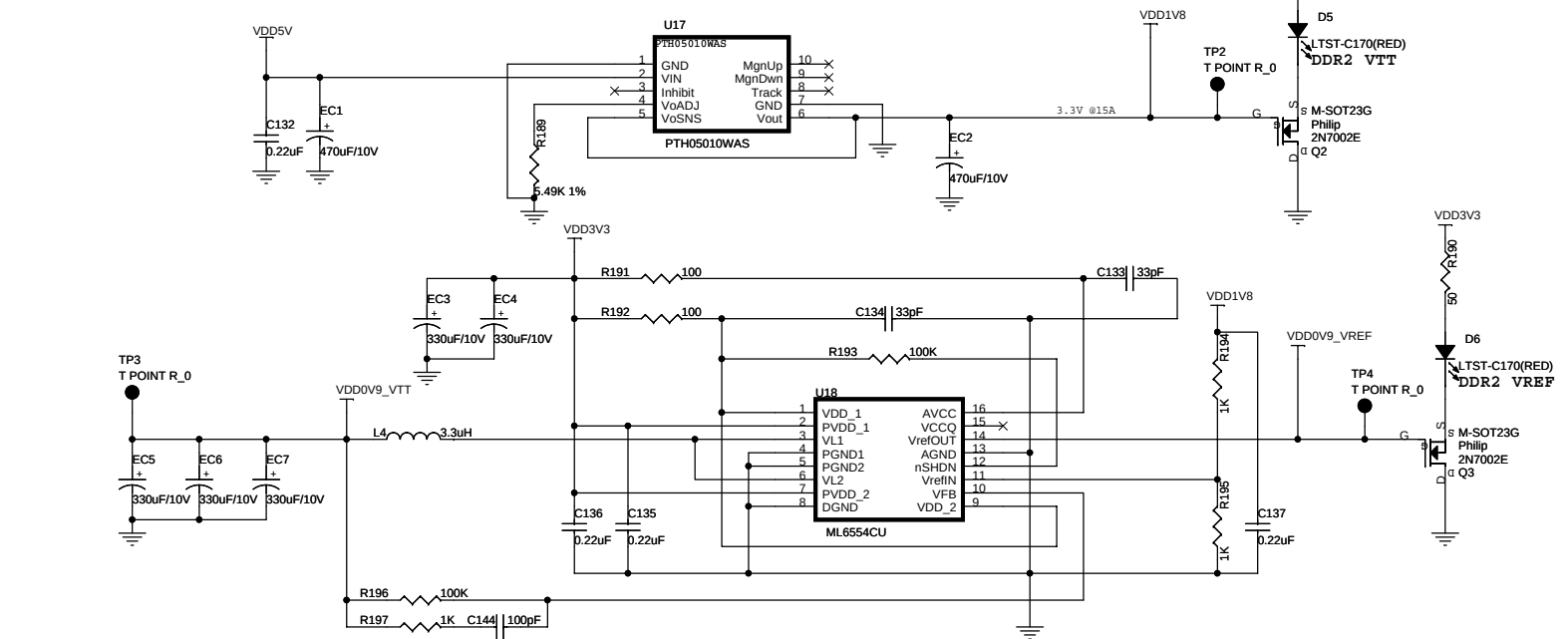
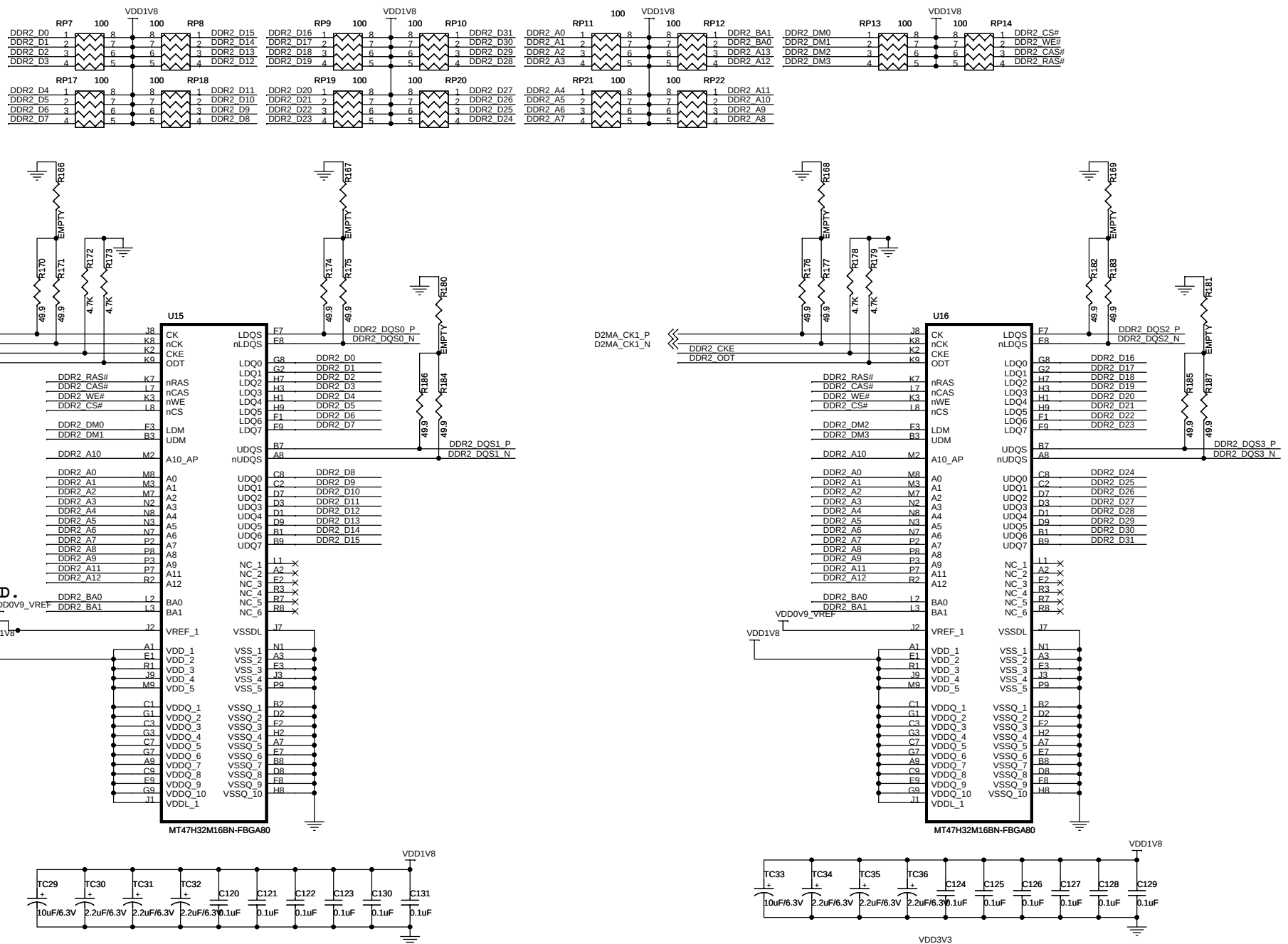
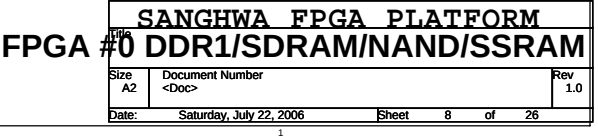
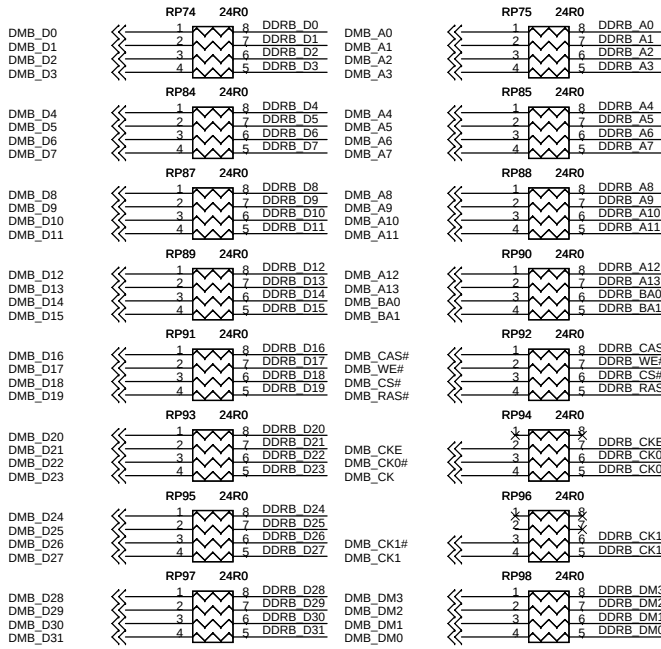


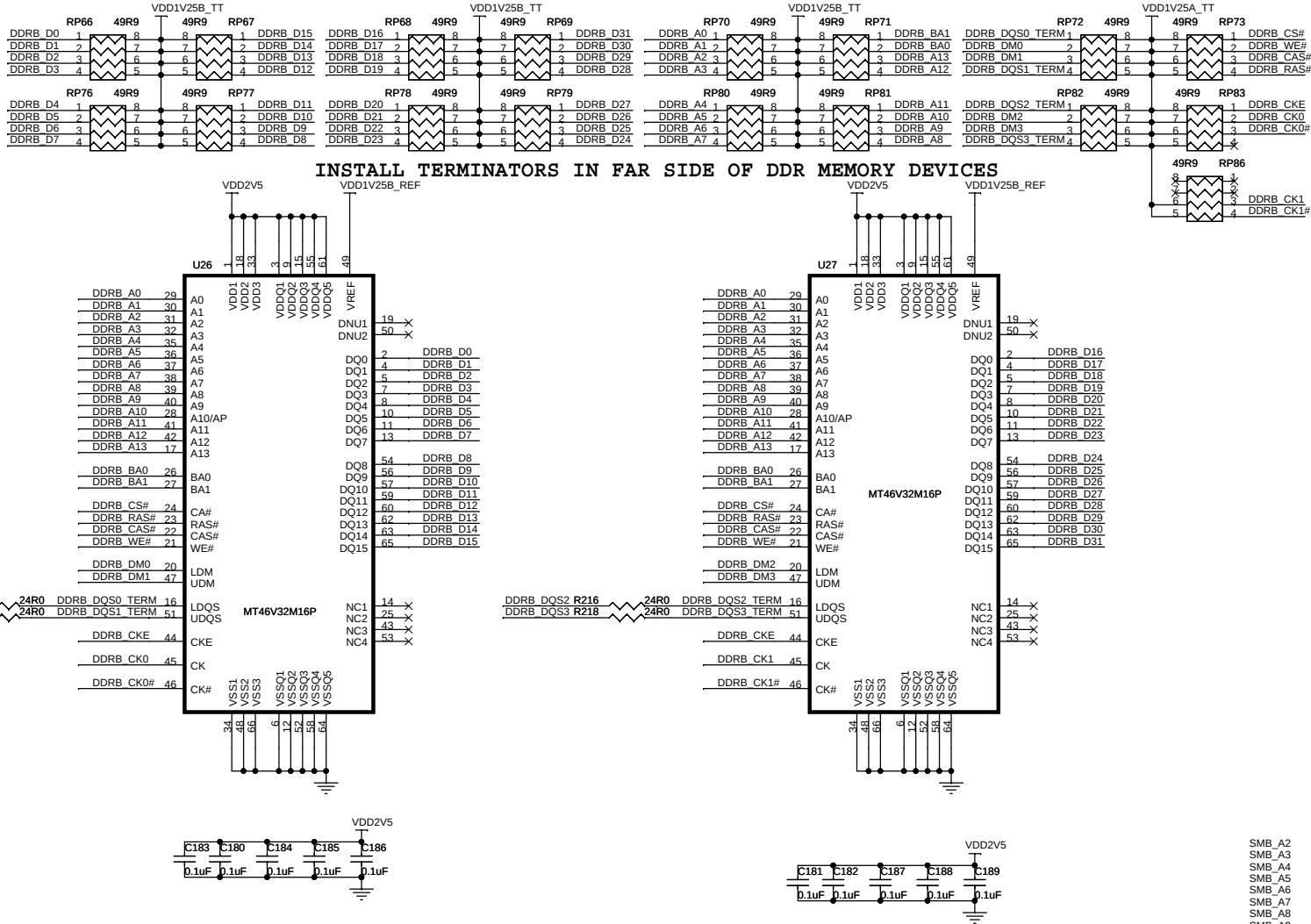
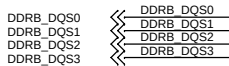
Figure 6-68: SSTL (1.8V) Class II Termination



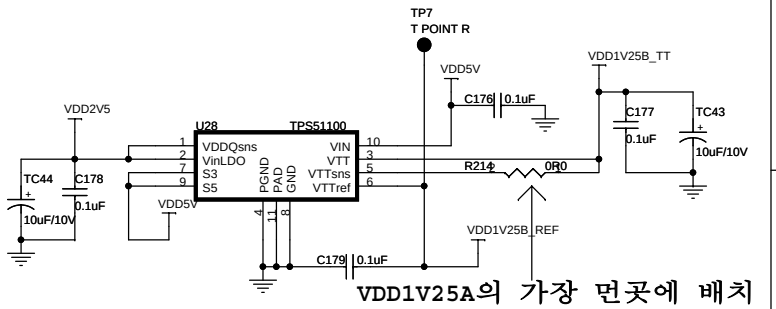




PLACE CLOSE TO DDR MEMORY DEVICES

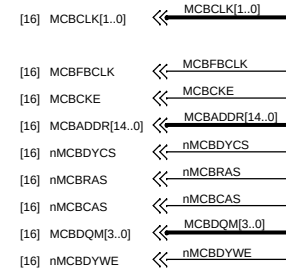


INSTALL TERMINATORS IN FAR SIDE OF DDR MEMORY DEVICES

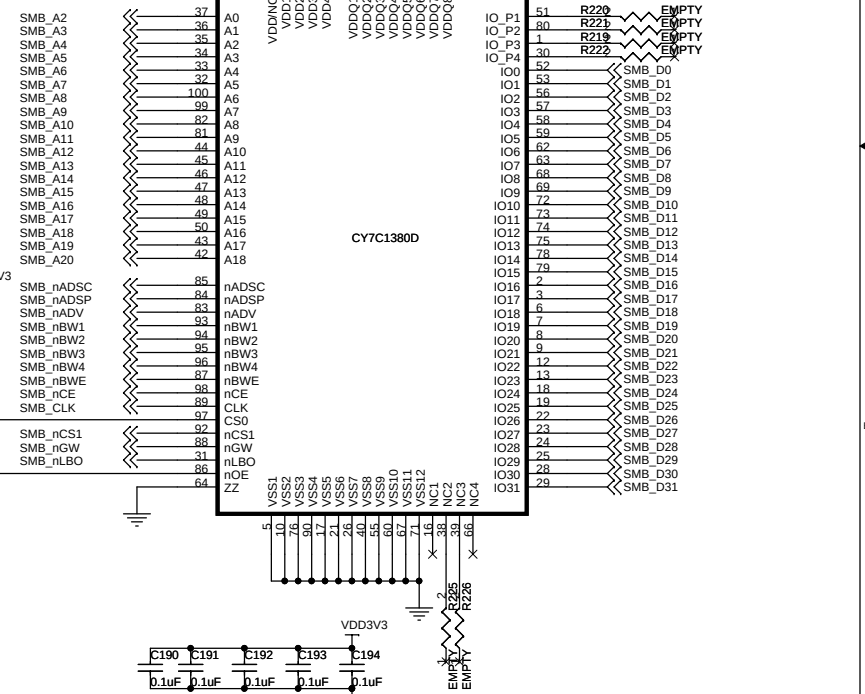
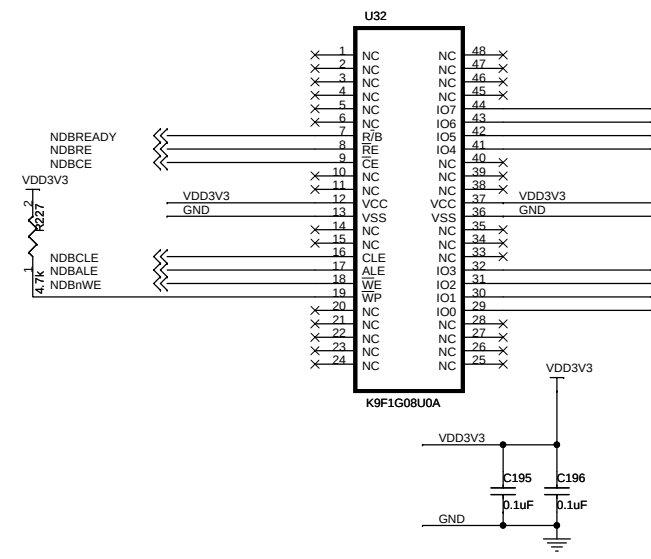
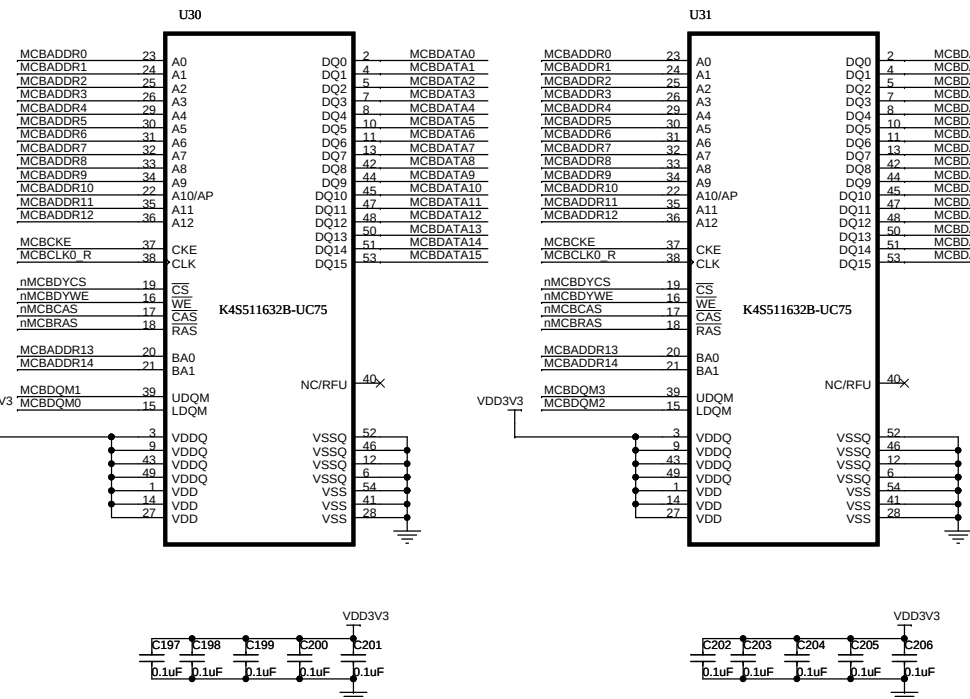


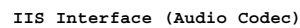
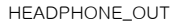
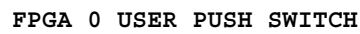
VDD1V25A의 가장 먼곳에 배치

DYNAMIC MEMORY INTERFACE

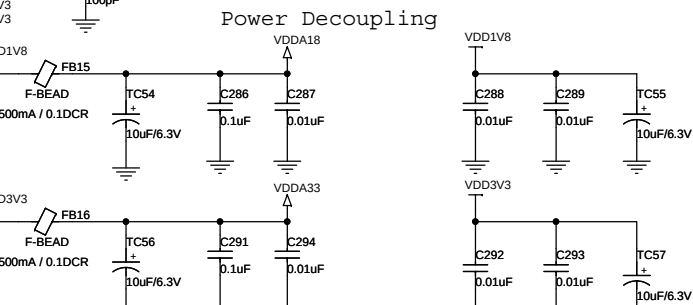
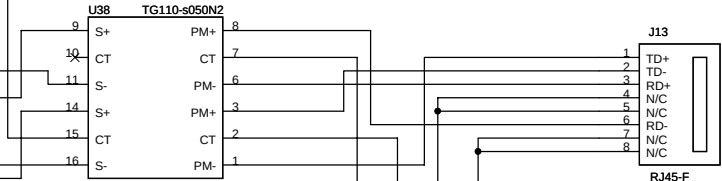
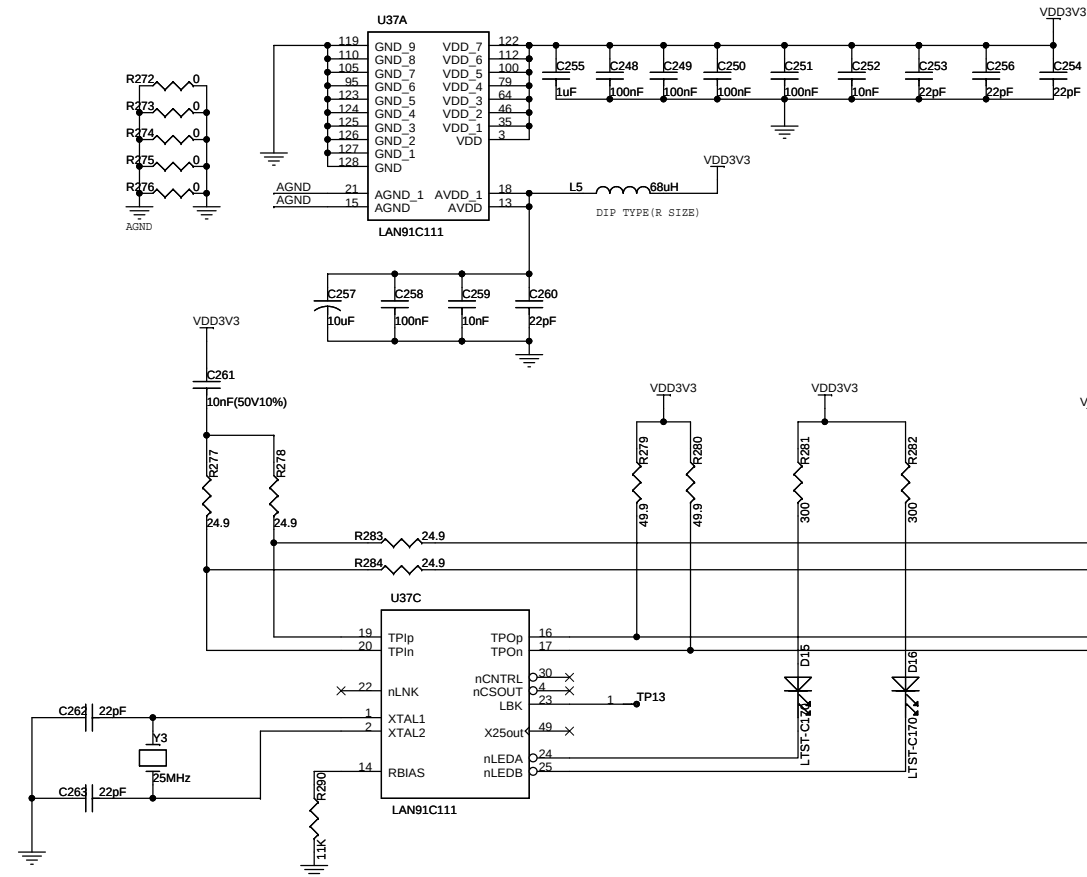


SDRAM(128MB) INTERFACE



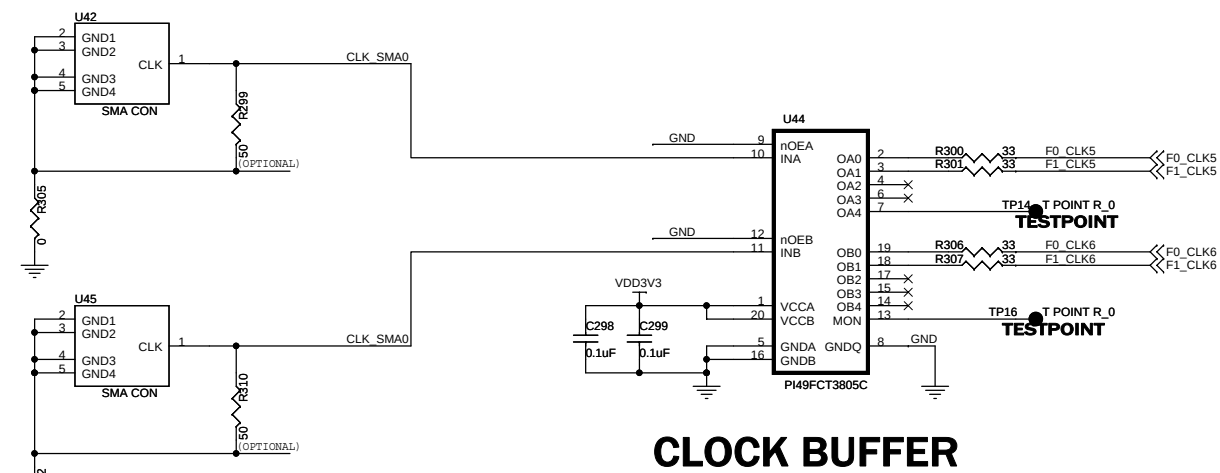
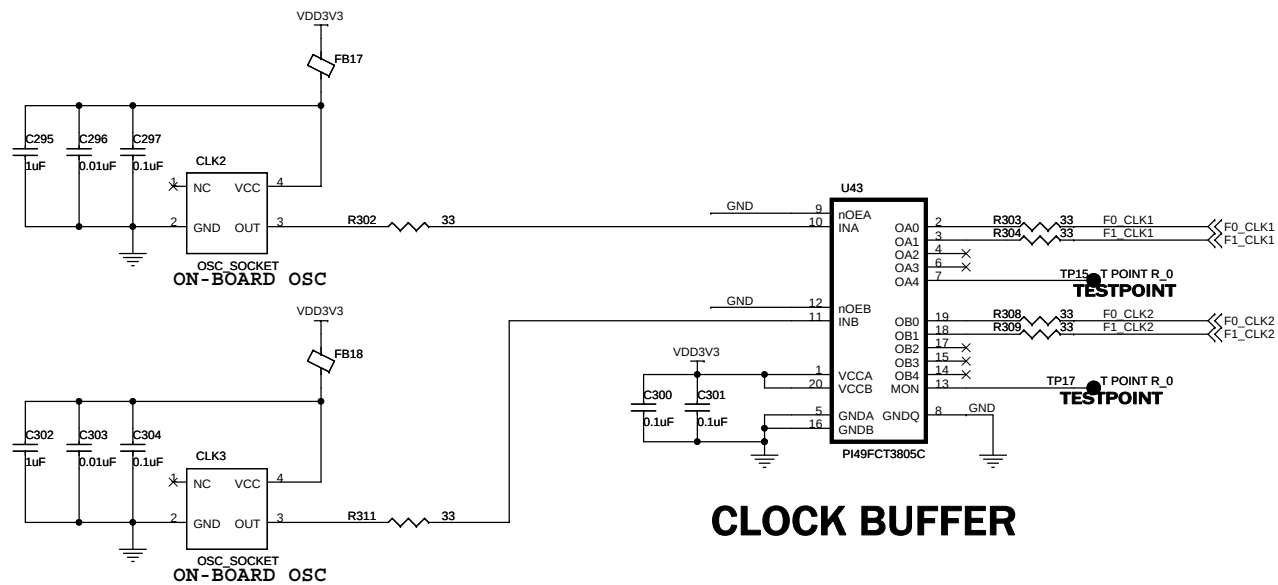


ETH_INTR	ETH_INTR
ETH_CS	ETH_CS
ETH_RESET	ETH_RESET
ETH_LCLK	ETH_LCLK



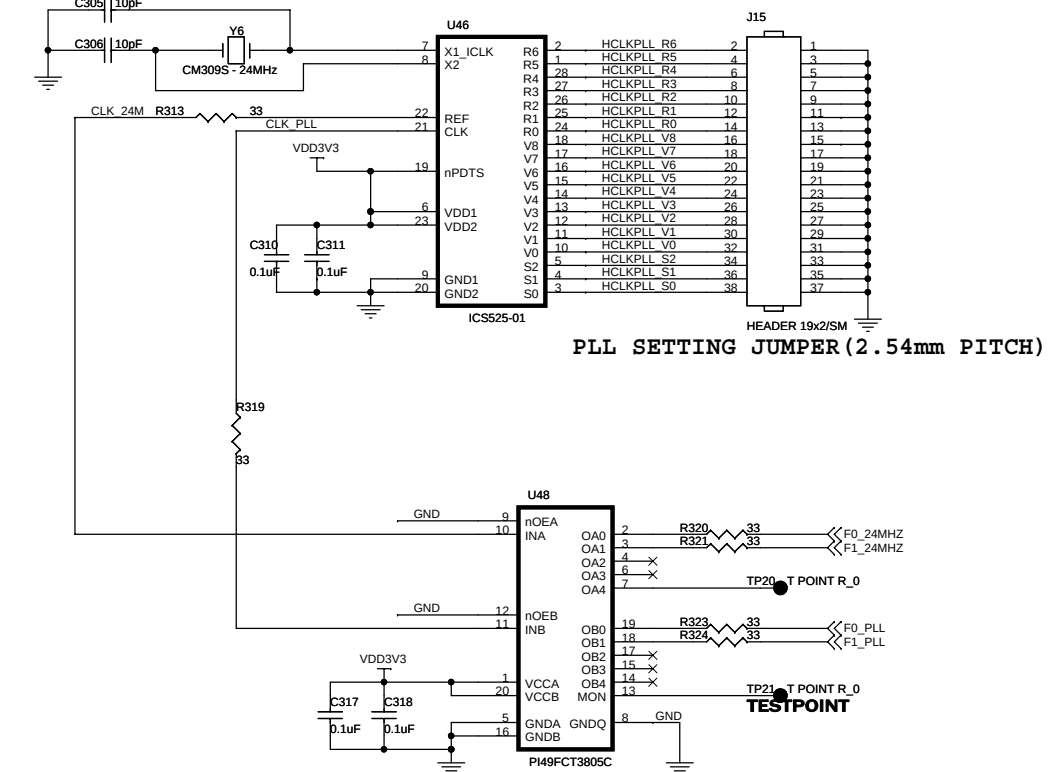
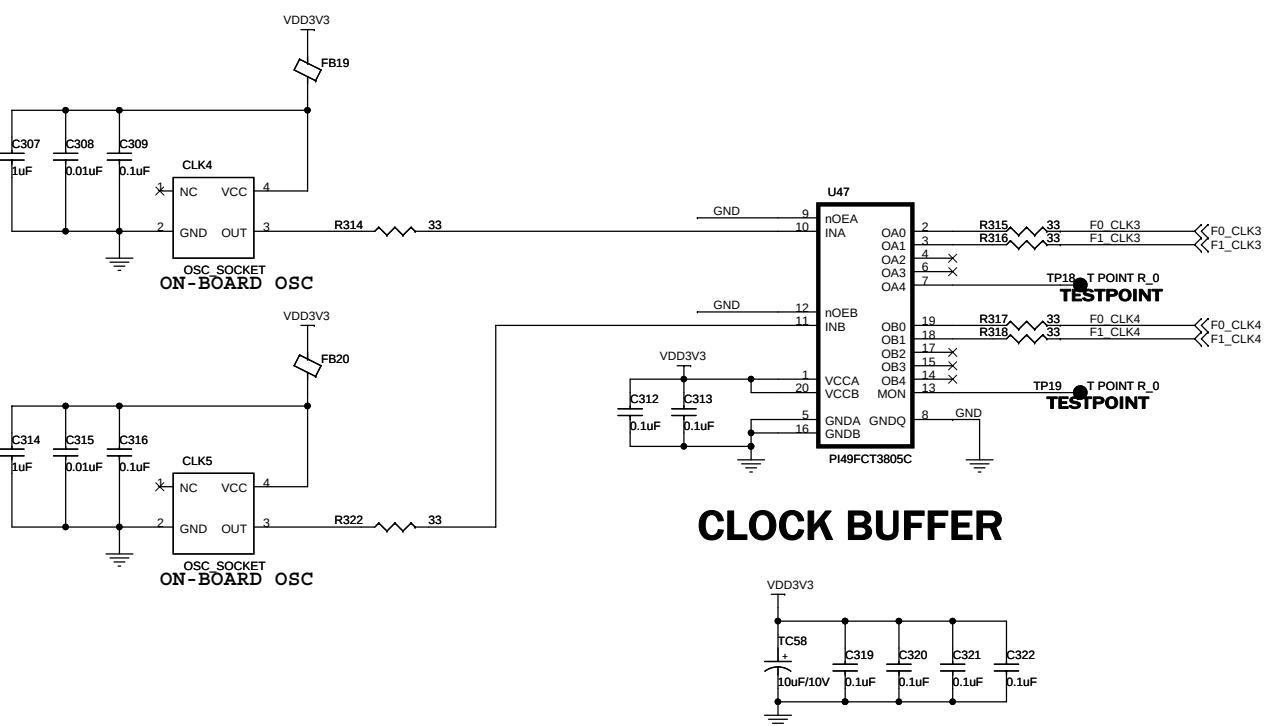
Title **SANGHWA FPGA PLATFORM**
ETH/USB/UART

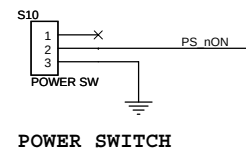
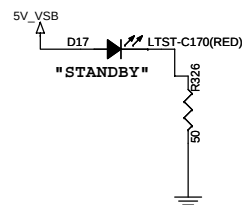
Date:	Saturday, July 22, 2006	Sheet	11	of	26
-------	-------------------------	-------	----	----	----



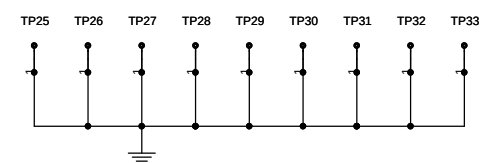
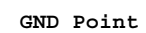
2006-3-4 부품 제거

SW10, SW11, SW12

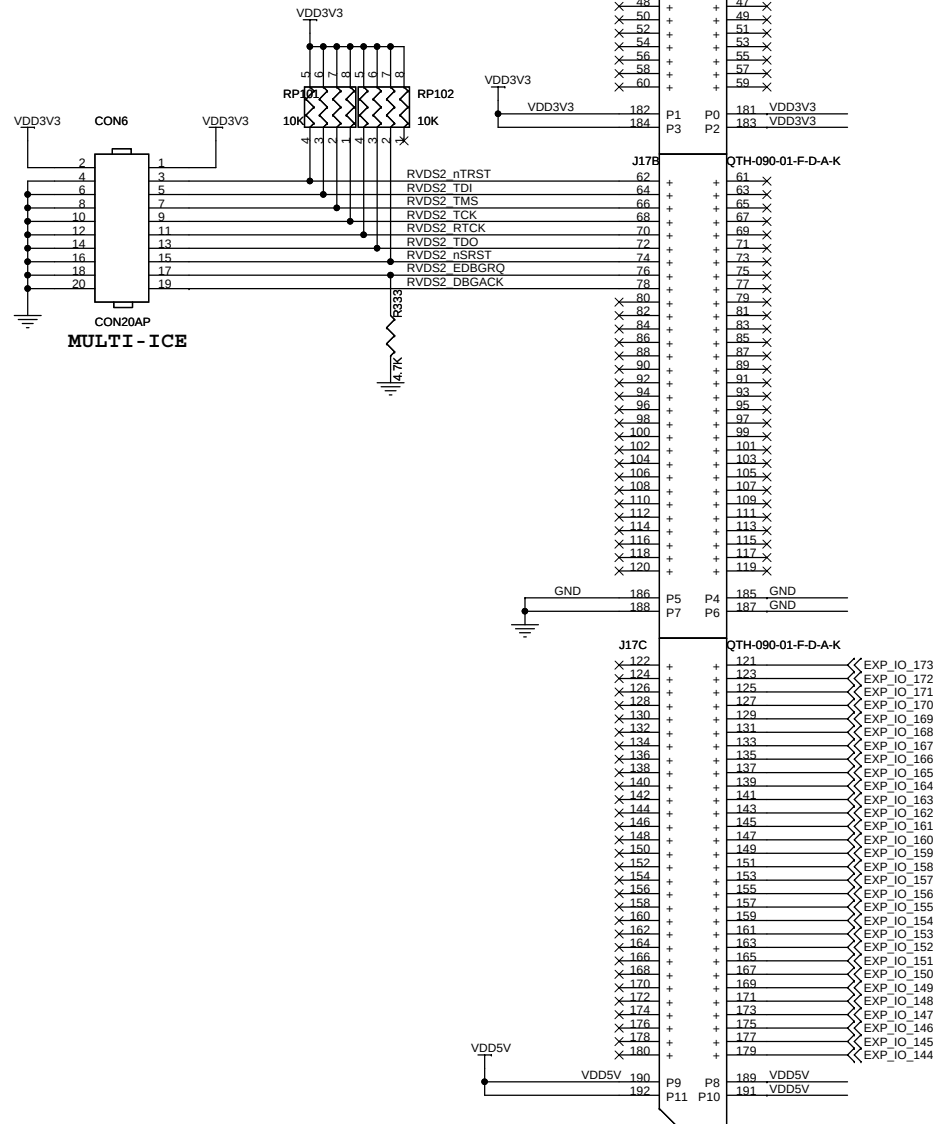
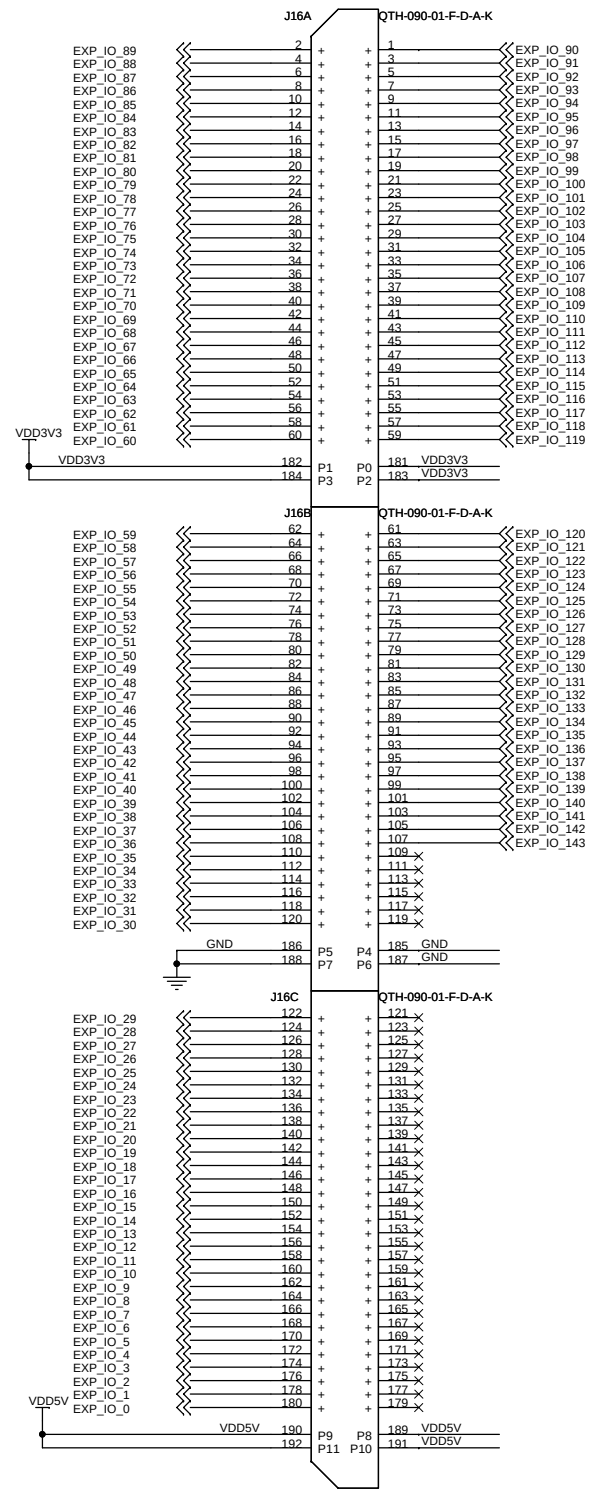


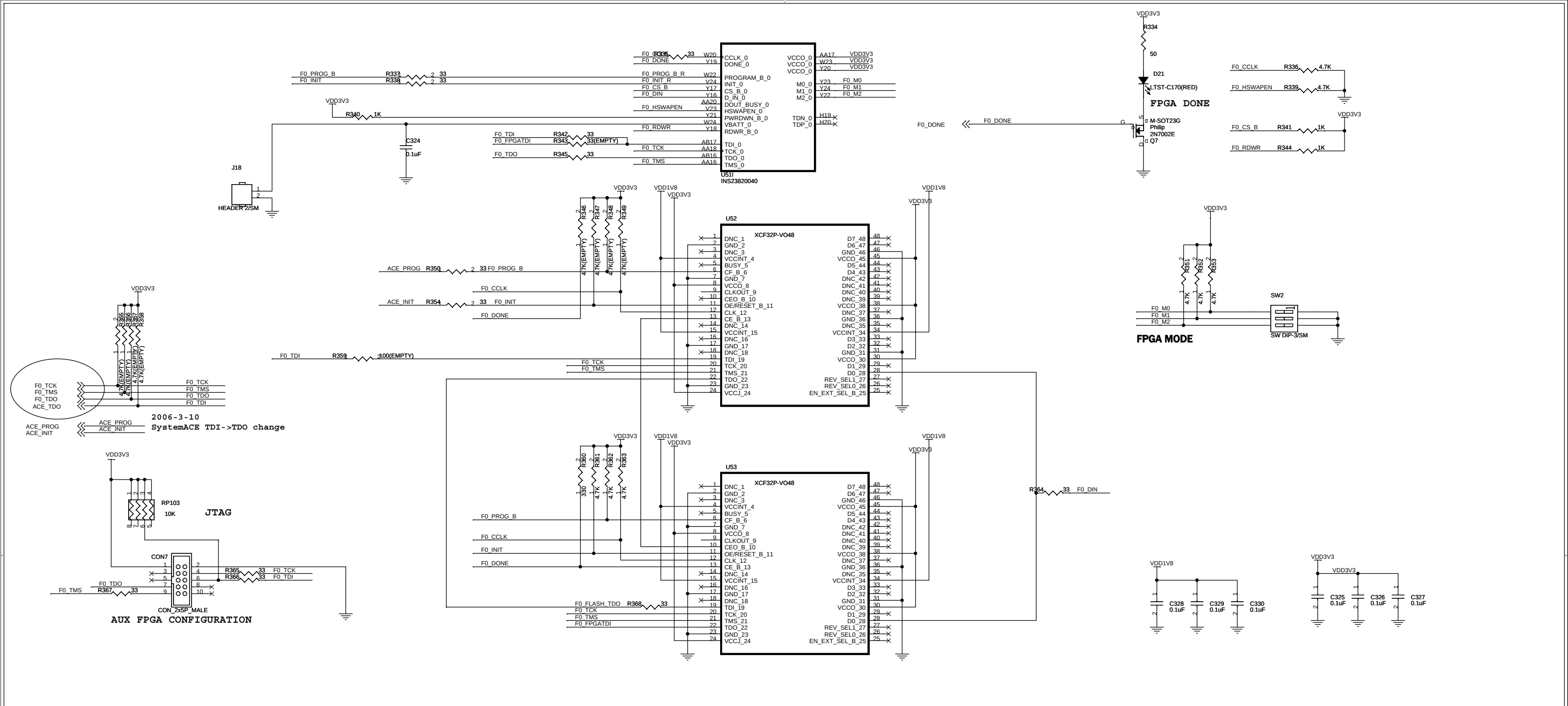


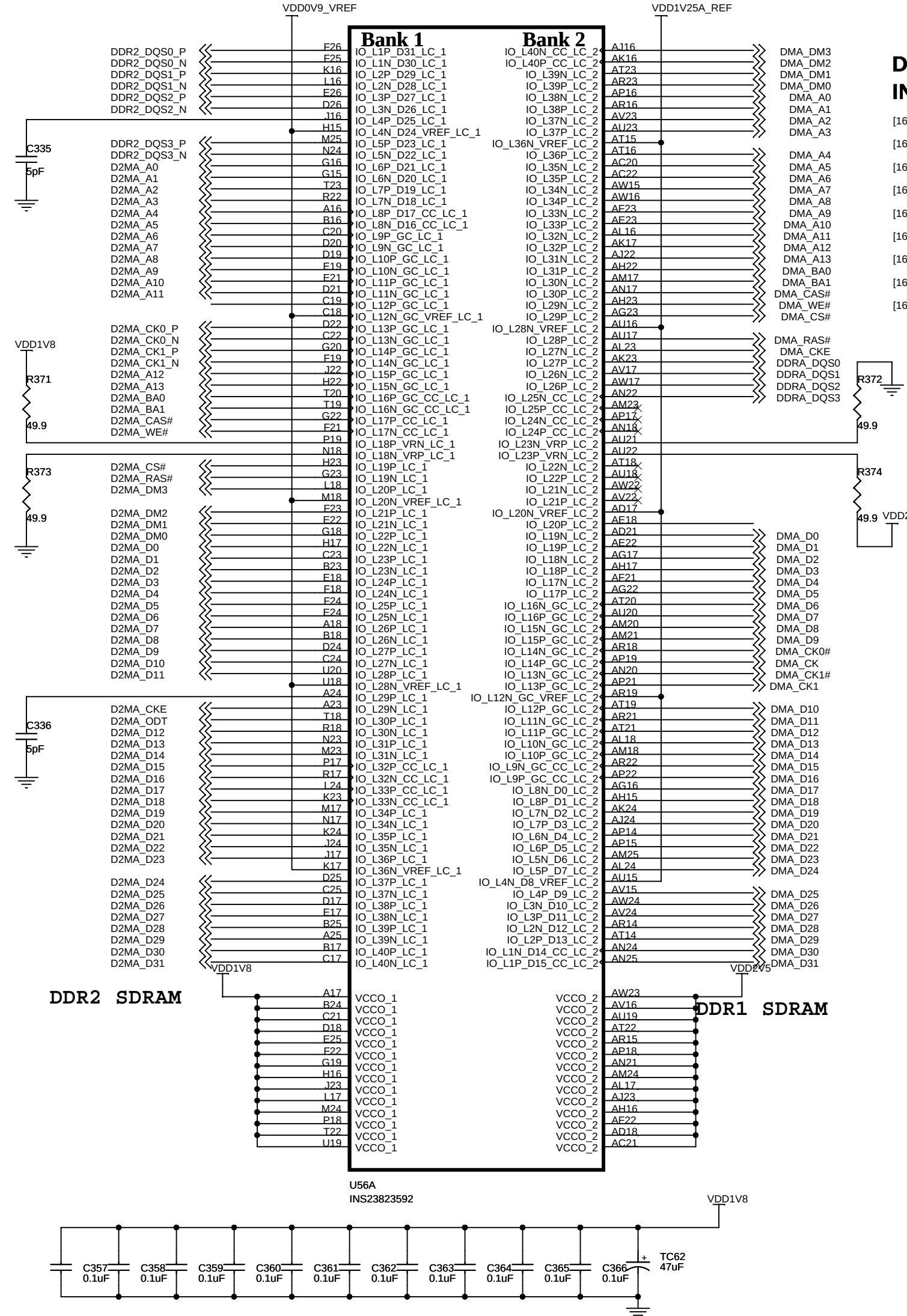
부품 디멘전에 주의 요망!!!!



EXPANSION SLOT FOR FPGA #0

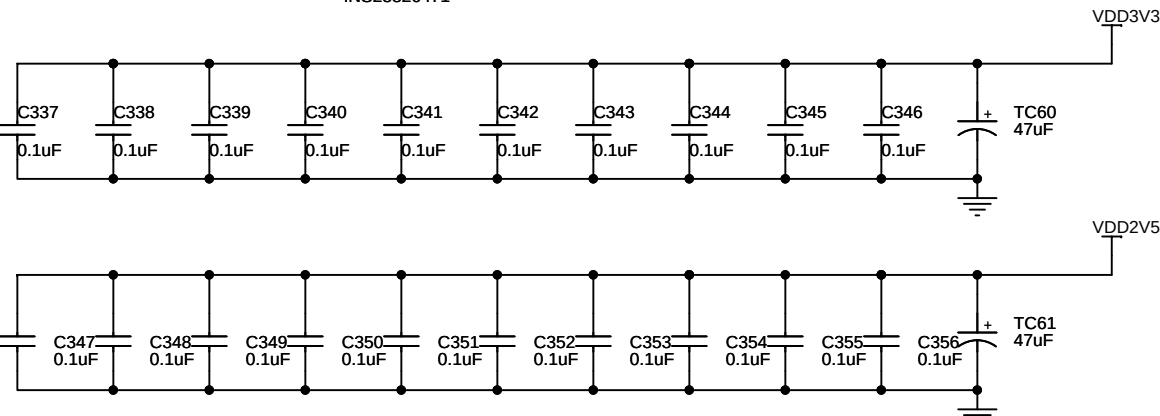
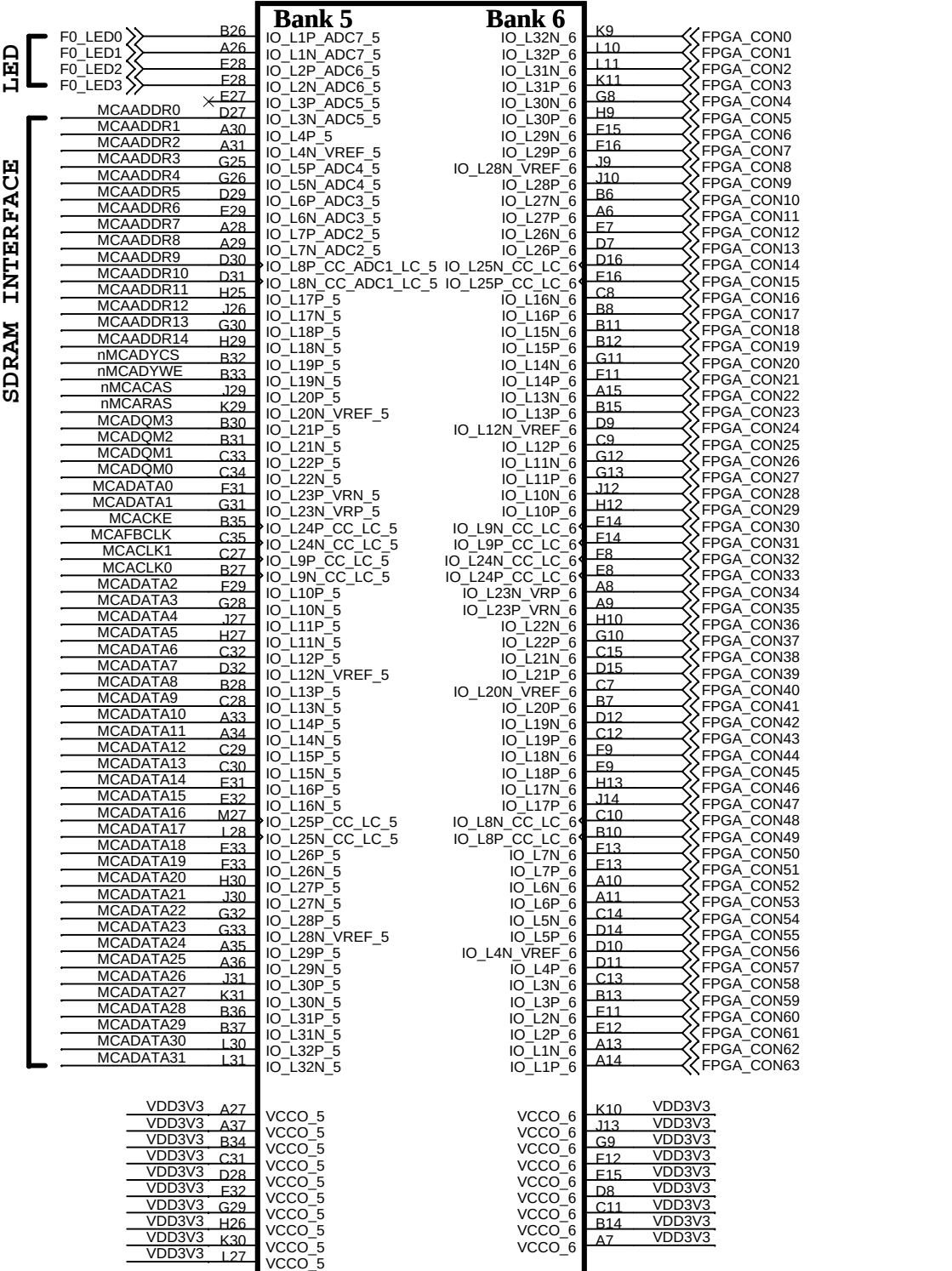






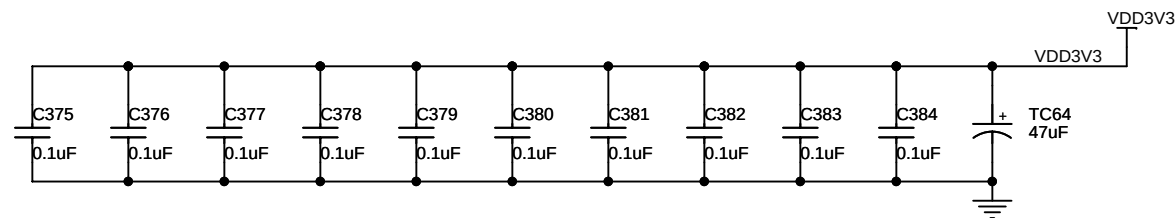
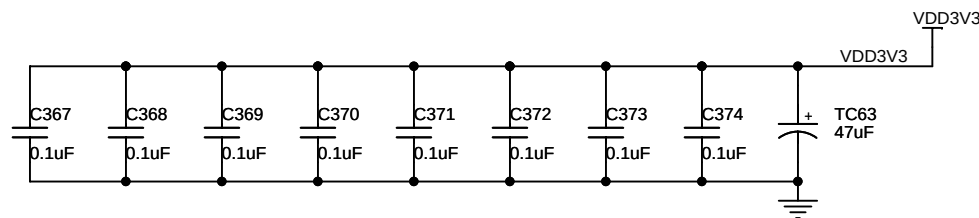
DYNAMIC MEMORY INTERFACE

- [16] MCACLK[1..0] << MCACLK[1..0]
- [16] MCAFBCLK << MCAFBCLK
- [16] MCACKE << MCACKE
- [16] MCAADDR[14..0] << MCAADDR[14..0]
- [16] nMCADYCS << nMCADYCS
- [16] nMCARAS << nMCARAS
- [16] nMCACAS << nMCACAS
- [16] MCADQM[3..0] << MCADQM[3..0]
- [16] nMCADYWE << nMCADYWE



FPGA INTERCONNECTION

Bank 7			Bank 8			Bank 9			Bank 10		
FPGA_CON384	AT33	IO_L1P_7	IO_L32N_8	AW10	FPGA_CON320	EXP_IO_64	D34	IO_L1P_9	IO_L32N_10	T11	FPGA_CON64
FPGA_CON385	AR33	IO_L1N_7	IO_L32P_8	AW11	FPGA_CON321	EXP_IO_65	D35	IO_L32P_9	IO_L32P_10	U12	FPGA_CON65
FPGA_CON386	AJ30	IO_L2P_7	IO_L31N_8	AV14	FPGA_CON322	EXP_IO_66	C37	IO_L2P_9	IO_L31N_10	R8	FPGA_CON66
FPGA_CON387	AK31	IO_L2N_7	IO_L31P_8	AW14	FPGA_CON323	EXP_IO_67	C38	IO_L2N_9	IO_L31P_10	R9	FPGA_CON67
FPGA_CON388	AM30	IO_L3P_7	IO_L30N_8	AU10	FPGA_CON324	EXP_IO_68	F34	IO_L3P_9	IO_L30N_10	P7	FPGA_CON68
FPGA_CON389	AL30	IO_L3N_7	IO_L30P_8	AV10	FPGA_CON325	EXP_IO_69	F34	IO_L3N_9	IO_L30P_10	N7	FPGA_CON69
FPGA_CON390	AM31	IO_L4P_7	IO_L29N_8	AR13	FPGA_CON326	EXP_IO_70	F35	IO_L4P_9	IO_L29N_10	M6	FPGA_CON70
FPGA_CON391	AL31	IO_L4N_VREF_7	IO_L29P_8	AT13	FPGA_CON327	EXP_IO_71	G35	IO_L4N_VREF_9	IO_L29P_10	L6	FPGA_CON71
FPGA_CON392	AP24	IO_L5P_7	IO_L28N_VREF_8	AR11	FPGA_CON328	EXP_IO_72	D36	IO_L5P_9	IO_L28N_VREF_10	N8	FPGA_CON72
FPGA_CON393	AR24	IO_L5N_7	IO_L28P_8	AT11	FPGA_CON329	EXP_IO_73	D37	IO_L5N_9	IO_L28P_10	P9	FPGA_CON73
FPGA_CON394	AP32	IO_L6P_7	IO_L27N_8	AM13	FPGA_CON330	EXP_IO_74	H33	IO_L6P_9	IO_L27N_10	M7	FPGA_CON74
FPGA_CON395	AN32	IO_L6N_7	IO_L27P_8	AN14	FPGA_CON331	EXP_IO_75	H34	IO_L6N_9	IO_L27P_10	M8	FPGA_CON75
FPGA_CON396	AN30	IO_L7P_7	IO_L26N_8	AV9	FPGA_CON332	EXP_IO_76	F36	IO_L7P_9	IO_L26N_10	U13	FPGA_CON76
FPGA_CON397	AP31	IO_L7N_7	IO_L26P_8	AW9	FPGA_CON333	EXP_IO_77	F36	IO_L7N_9	IO_L26P_10	T13	FPGA_CON77
FPGA_CON398	AK29	IO_L8P_CC_LC_7	IO_L25N_CC_LC_8	AV12	FPGA_CON334	EXP_IO_78	C39	IO_L8P_CC_LC_9	IO_L25N_CC_LC_10	R11	FPGA_CON78
FPGA_CON399	AJ29	IO_L8N_CC_LC_7	IO_L25P_CC_LC_8	AW12	FPGA_CON335	EXP_IO_79	D39	IO_L8N_CC_LC_9	IO_L25P_CC_LC_10	P11	FPGA_CON79
FPGA_CON400	AT24	IO_L9P_CC_LC_7	IO_L24N_CC_LC_8	AV7	FPGA_CON336	EXP_IO_80	J32	IO_L9P_CC_LC_9	IO_L24N_CC_LC_10	G1	FPGA_CON80
FPGA_CON401	AT25	IO_L9N_CC_LC_7	IO_L24P_CC_LC_8	AV8	FPGA_CON337	EXP_IO_81	K32	IO_L9N_CC_LC_9	IO_L24P_CC_LC_10	F1	FPGA_CON81
FPGA_CON402	AW34	IO_L10P_7	IO_L23N_VRP_8	AU11	FPGA_CON338	EXP_IO_82	G36	IO_L10P_9	IO_L23N_VRP_10	K6	FPGA_CON82
FPGA_CON403	AV34	IO_L10N_7	IO_L23P_VRN_8	AU12	FPGA_CON339	EXP_IO_83	G37	IO_L10N_9	IO_L23P_VRN_10	J6	FPGA_CON83
FPGA_CON404	AW30	IO_L11P_7	IO_L22N_8	AT9	FPGA_CON340	EXP_IO_84	F37	IO_L11P_9	IO_L22N_10	G2	FPGA_CON84
FPGA_CON405	AW31	IO_L11N_7	IO_L22P_8	AT10	FPGA_CON341	EXP_IO_85	F38	IO_L11N_9	IO_L22P_10	G3	FPGA_CON85
FPGA_CON406	AV33	IO_L12P_7	IO_L21N_8	AU13	FPGA_CON342	EXP_IO_86	H35	IO_L12P_9	IO_L21N_10	J5	FPGA_CON86
FPGA_CON407	AU33	IO_L12N_VREF_7	IO_L21P_8	AV13	FPGA_CON343	EXP_IO_87	J35	IO_L12N_VREF_9	IO_L21P_10	H5	FPGA_CON87
FPGA_CON408	AP25	IO_L13P_7	IO_L20N_VREF_8	AP9	FPGA_CON344	EXP_IO_88	M30	IO_L13P_9	IO_L20N_VREF_10	L8	FPGA_CON88
FPGA_CON409	AP26	IO_L13N_7	IO_L20P_8	AR9	FPGA_CON345	EXP_IO_89	M31	IO_L13N_9	IO_L20P_10	K7	FPGA_CON89
FPGA_CON410	AT31	IO_L14P_7	IO_L19N_8	AP12	FPGA_CON346	EXP_IO_90	F39	IO_L14P_9	IO_L19N_10	R12	FPGA_CON90
FPGA_CON411	AT30	IO_L14N_7	IO_L19P_8	AR12	FPGA_CON347	EXP_IO_91	F39	IO_L14N_9	IO_L19P_10	P12	FPGA_CON91
FPGA_CON412	AU25	IO_L15P_7	IO_L18N_8	AT8	FPGA_CON348	EXP_IO_92	J34	IO_L15P_9	IO_L18N_10	E1	FPGA_CON92
FPGA_CON413	AV25	IO_L15N_7	IO_L18P_8	AU8	FPGA_CON349	EXP_IO_93	K34	IO_L15N_9	IO_L18P_10	E2	FPGA_CON93
FPGA_CON414	AR31	IO_L16P_7	IO_L17N_8	AL13	FPGA_CON350	EXP_IO_94	F38	IO_L16P_9	IO_L17N_10	T14	FPGA_CON94
FPGA_CON415	AR32	IO_L16N_7	IO_L17P_8	AL14	FPGA_CON351	EXP_IO_95	G38	IO_L16N_9	IO_L17P_10	R14	FPGA_CON95
FPGA_CON416	AI26	IO_L17P_7	IO_L16N_8	AU6	FPGA_CON352	EXP_IO_96	K33	IO_L17P_9	IO_L16N_10	D1	FPGA_CON96
FPGA_CON417	AM27	IO_L17N_7	IO_L16P_8	AU7	FPGA_CON353	EXP_IO_97	L33	IO_L17N_9	IO_L16P_10	D2	FPGA_CON97
FPGA_CON418	AU31	IO_L18P_7	IO_L15N_8	AP11	FPGA_CON354	EXP_IO_98	H37	IO_L18P_9	IO_L15N_10	E3	FPGA_CON98
FPGA_CON419	AU32	IO_L18N_7	IO_L15P_8	AN12	FPGA_CON355	EXP_IO_99	H38	IO_L18N_9	IO_L15P_10	F4	FPGA_CON99
FPGA_CON420	AV28	IO_L19P_7	IO_L14N_8	AJ12	FPGA_CON356	EXP_IO_100	N30	IO_L19P_9	IO_L14N_10	N10	FPGA_CON100
FPGA_CON421	AU28	IO_L19N_7	IO_L14P_8	AH13	FPGA_CON357	EXP_IO_101	P29	IO_L19N_9	IO_L14P_10	M10	FPGA_CON101
FPGA_CON422	AW32	IO_L20P_7	IO_L13N_8	AN10	FPGA_CON358	EXP_IO_102	L34	IO_L20P_9	IO_L13N_10	T15	FPGA_CON102
FPGA_CON423	AV32	IO_L20N_VREF_7	IO_L13P_8	AM11	FPGA_CON359	EXP_IO_103	L35	IO_L20N_VREF_9	IO_L13P_10	R16	FPGA_CON103
FPGA_CON424	AW25	IO_L21P_7	IO_L12N_VREF_8	AI10	FPGA_CON360	EXP_IO_104	J36	IO_L21P_9	IO_L12N_VREF_10	E3	FPGA_CON104
FPGA_CON425	AW26	IO_L21N_7	IO_L12P_8	AM10	FPGA_CON361	EXP_IO_105	J37	IO_L21N_9	IO_L12P_10	F4	FPGA_CON105
FPGA_CON426	AP29	IO_L22P_7	IO_L11N_8	AW6	FPGA_CON362	EXP_IO_106	K36	IO_L22P_9	IO_L11N_10	N12	FPGA_CON106
FPGA_CON427	AN29	IO_L22N_7	IO_L11P_8	AW7	FPGA_CON363	EXP_IO_107	L36	IO_L22N_9	IO_L11P_10	M11	FPGA_CON107
FPGA_CON428	AN27	IO_L23P_VRN_7	IO_L10N_8	AW4	FPGA_CON364	EXP_IO_108	H39	IO_L23P_VRN_9	IO_L10N_10	G5	FPGA_CON108
FPGA_CON429	AN28	IO_L23N_VRP_7	IO_L10P_8	AW5	FPGA_CON365	EXP_IO_109	J39	IO_L23N_VRP_9	IO_L10P_10	E5	FPGA_CON109
FPGA_CON430	AL28	IO_L24P_CC_LC_7	IO_L9N_CC_LC_8	AK11	FPGA_CON366	EXP_IO_110	M33	IO_L24P_CC_LC_9	IO_L9N_CC_LC_10	G6	FPGA_CON110
FPGA_CON431	AM28	IO_L24N_CC_LC_7	IO_L9P_CC_LC_8	AL11	FPGA_CON367	EXP_IO_111	N32	IO_L24N_CC_LC_9	IO_L9P_CC_LC_10	G7	FPGA_CON111
FPGA_CON432	AP27	IO_L25P_CC_SM7_LC_7	IO_L8N_CC_LC_8	AV3	FPGA_CON368	EXP_IO_112	R28	IO_L25P_CC_LC_9	IO_L8N_CC_LC_10	J7	FPGA_CON112
FPGA_CON433	AR27	IO_L25N_CC_SM7_LC_7	IO_L8P_CC_LC_8	AV4	FPGA_CON369	EXP_IO_113	R29	IO_L25N_CC_LC_9	IO_L8P_CC_LC_10	H7	FPGA_CON113
FPGA_CON434	AV30	IO_L26P_SM6_7	IO_L7N_8	AR7	FPGA_CON370	EXP_IO_114	M35	IO_L26P_9	IO_L7N_10	F6	FPGA_CON114
FPGA_CON435	AU30	IO_L26N_SM6_7	IO_L7P_8	AR8	FPGA_CON371	EXP_IO_115	N35	IO_L26N_9	IO_L7P_10	F6	FPGA_CON115
FPGA_CON436	AR26	IO_L27P_SM5_7	IO_L6N_8	AN7	FPGA_CON372	EXP_IO_116	K37	IO_L27P_9	IO_L6N_10	C2	FPGA_CON116
FPGA_CON437	AT26	IO_L27N_SM5_7	IO_L6P_8	AP7	FPGA_CON373	EXP_IO_117	K38	IO_L27N_9	IO_L6P_10	C3	FPGA_CON117
FPGA_CON438	AW29	IO_L28P_7	IO_L5N_8	AN15	FPGA_CON374	EXP_IO_118	N33	IO_L28P_9	IO_L5N_10	D5	FPGA_CON118
FPGA_CON439	AV29	IO_L28N_VREF_7	IO_L5P_8	AM15	FPGA_CON375	EXP_IO_119	N34	IO_L28N_VREF_9	IO_L5P_10	D6	FPGA_CON119
FPGA_CON440	AV27	IO_L29P_SM4_7	IO_L4N_VREF_8	AK9	FPGA_CON376	EXP_IO_120	P31	IO_L29P_9	IO_L4N_VREF_10	D4	FPGA_CON120
FPGA_CON441	AW27	IO_L29N_SM4_7	IO_L4P_8	AL9	FPGA_CON377	EXP_IO_121	P32	IO_L29N_9	IO_L4P_10	C4	FPGA_CON121
FPGA_CON442	AT29	IO_L30P_SM3_7	IO_L3N_8	AN8	FPGA_CON378	EXP_IO_122	K39	IO_L30P_9	IO_L3N_10	C5	FPGA_CON122
FPGA_CON443	AR29	IO_L30N_SM3_7	IO_L3P_8	AN9	FPGA_CON379	EXP_IO_123	L39	IO_L30N_9	IO_L3P_10	B5	FPGA_CON123
FPGA_CON444	AU26	IO_L31P_SM2_7	IO_L2N_8	AJ9	FPGA_CON380	EXP_IO_124	L38	IO_L31P_9	IO_L2N_10	B3	FPGA_CON124
FPGA_CON445	AU27	IO_L31N_SM2_7	IO_L2P_8	AI10	FPGA_CON381	EXP_IO_125	M38	IO_L31N_9	IO_L2P_10	A3	FPGA_CON125
FPGA_CON446	AT28	IO_L32P_SM1_7	IO_L1N_8	AU5	FPGA_CON382	EXP_IO_126	M36	IO_L32P_9	IO_L1N_10	A4	FPGA_CON126
FPGA_CON447	AR28	IO_L32N_SM1_7	IO_L1P_8	AV5	FPGA_CON383	EXP_IO_127	M37	IO_L32N_9	IO_L1P_10	A5	FPGA_CON127



FPGA INTERCONNECTION

SMA_nOE

SSRAM INTERFACE

NAND FLASH

NDAREADY
NDARE
NDACE
NDACLE
NDAALE
NDAAnWE
NDA_D7
NDA_D6
NDA_D5
NDA_D4
NDA_D3
NDA_D2
NDA_D1
NDA_D0

EXP_IO_173
EXP_IO_172
EXP_IO_171
EXP_IO_170
EXP_IO_169
EXP_IO_168
EXP_IO_167
EXP_IO_166
EXP_IO_165
EXP_IO_164
EXP_IO_163
EXP_IO_162
EXP_IO_161
EXP_IO_160
EXP_IO_159
EXP_IO_158
EXP_IO_157
EXP_IO_156
EXP_IO_155
EXP_IO_154
EXP_IO_153
EXP_IO_152
EXP_IO_151
EXP_IO_150
EXP_IO_149
EXP_IO_148
EXP_IO_147
EXP_IO_146
EXP_IO_145
EXP_IO_144
EXP_IO_143
EXP_IO_142
EXP_IO_141
EXP_IO_140
EXP_IO_139
EXP_IO_138
EXP_IO_137
EXP_IO_136
EXP_IO_135
EXP_IO_134
EXP_IO_133
EXP_IO_132
EXP_IO_131
EXP_IO_130
EXP_IO_129
EXP_IO_128

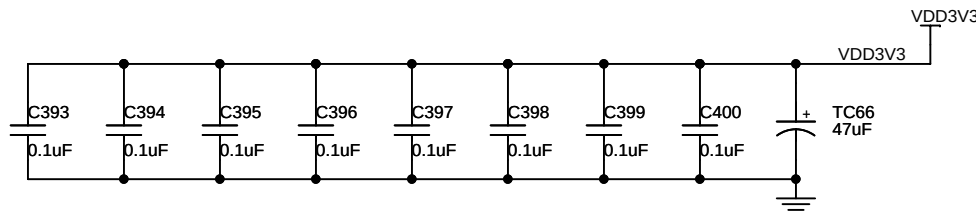
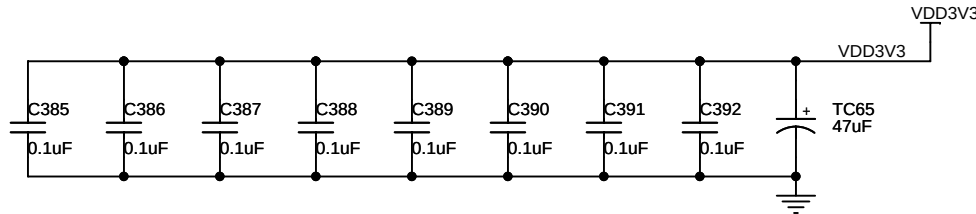
VDD3V3 AD28
VDD3V3 AE25
VDD3V3 AF32
VDD3V3 AG29
VDD3V3 AI33
VDD3V3 AM34
VDD3V3 AP38
VDD3V3 AR35
VDD3V3 AU39
VDD3V3 AV36

VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11

VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12

AT2 VDD3V3
AR5 VDD3V3
AN1 VDD3V3
AM4 VDD3V3
AL7 VDD3V3
AJ3 VDD3V3
AH6 VDD3V3
AG9 VDD3V3
AF12 VDD3V3

U59F
INS23839244



SSRAM INTERFACE

VDD3V3 AA37
VDD3V3 P38
VDD3V3 R35
VDD3V3 T32
VDD3V3 U29
VDD3V3 U36
VDD3V3 V26
VDD3V3 V36
VDD3V3 W33

VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13

U60G
INS23841562



VIRTEX-4 FPGA
POWER GROUND

GND D13	GND	GND J28	GND
GND K13	GND	GND N28	GND
GND M13	GND	GND W28	GND
GND P13	GND	GND AJ28	GND
GND AD13	GND	GND AW28	GND
GND AE13	GND	GND B29	GND
GND AK13	GND	GND M29	GND
GND AP13	GND	GND AB29	GND
GND G14	GND	GND AM29	GND
GND I14	GND	GND E30	GND
GND N14	GND	GND R30	GND
GND U14	GND	GND AE30	GND
GND W14	GND	GND AR30	GND
GND AE14	GND	GND H31	GND
GND AG14	GND	GND V31	GND
GND AJ14	GND	GND AH31	GND
GND AU14	GND	GND AV31	GND
GND K15	GND	GND A32	GND
GND M15	GND	GND L32	GND
GND P15	GND	GND AA32	GND
GND V15	GND	GND AL32	GND
GND Y15	GND	GND D33	GND
GND AD15	GND	GND P33	GND
GND AE15	GND	GND AD33	GND
GND AK15	GND	GND AP33	GND
GND C16	GND	GND G34	GND
GND N16	GND	GND U34	GND
GND U16	GND	GND AG34	GND
GND W16	GND	GND AU34	GND
GND AC16	GND	GND K35	GND
GND AF16	GND	GND Y35	GND
GND AN16	GND	GND AK35	GND
GND F17	GND	GND C36	GND
GND T17	GND	GND N36	GND
GND V17	GND	GND AC36	GND
GND AF17	GND	GND AN36	GND
GND AT17	GND	GND E37	GND
GND J18	GND	GND T37	GND
GND W18	GND	GND AE37	GND
GND AC18	GND	GND AT37	GND
GND AJ18	GND	GND A38	GND
GND AW18	GND	GND J38	GND
GND B19	GND	GND W38	GND
GND M19	GND	GND AJ38	GND
GND V19	GND	GND AW38	GND
GND AB19	GND	GND B39	GND
GND AD19	GND	GND M39	GND
GND AM19	GND	GND AB39	GND
GND F20	GND	GND AM39	GND
GND R20	GND	GND AV39	GND

GND AE20	GND	GND B1	GND
GND AR20	GND	GND H1	GND
GND H21	GND	GND V1	GND
GND T21	GND	GND AH1	GND
GND V21	GND	GND AV1	GND
GND AB21	GND	GND A2	GND
GND AH21	GND	GND L2	GND
GND AV21	GND	GND AA2	GND
GND A22	GND	GND AL2	GND
GND I22	GND	GND AW2	GND
GND U22	GND	GND D3	GND
GND AA22	GND	GND P3	GND
GND AL22	GND	GND AD3	GND
GND D23	GND	GND AP3	GND
GND P23	GND	GND G4	GND
GND AB23	GND	GND U4	GND
GND AD23	GND	GND AG4	GND
GND AP23	GND	GND AU4	GND
GND G24	GND	GND K5	GND
GND R24	GND	GND Y5	GND
GND U24	GND	GND AK5	GND
GND AA24	GND	GND C6	GND
GND AC24	GND	GND N6	GND
GND AG24	GND	GND AC6	GND
GND AU24	GND	GND AN6	GND
GND K25	GND	GND F7	GND
GND P25	GND	GND T7	GND
GND T25	GND	GND AE7	GND
GND Y25	GND	GND AT7	GND
GND AB25	GND	GND J8	GND
GND AE25	GND	GND WR	GND
GND AH25	GND	GND AJ8	GND
GND AK25	GND	GND AW8	GND
GND C26	GND	GND B9	GND
GND I26	GND	GND M9	GND
GND N26	GND	GND AB9	GND
GND R26	GND	GND AM9	GND
GND AA26	GND	GND E10	GND
GND AC26	GND	GND R10	GND
GND AG26	GND	GND AE10	GND
GND AJ26	GND	GND AR10	GND
GND AN26	GND	GND H11	GND
GND F27	GND	GND V11	GND
GND K27	GND	GND AH11	GND
GND P27	GND	GND AV11	GND
GND T27	GND	GND A12	GND
GND AE27	GND	GND L12	GND
GND AH27	GND	GND AA12	GND
GND AK27	GND	GND AG12	GND
GND AT27	GND	GND AL12	GND

U61K
INS23847039

SSRAM INTERFACE

SMA_A2	
SMA_A3	
SMA_A4	
SMA_A5	
SMA_A6	
SMA_A7	
SMA_A8	
SMA_A9	
SMA_A10	
SMA_A11	
SMA_A12	
SMA_A13	
SMA_A14	
SMA_A15	
SMA_A16	
SMA_A17	
SMA_A18	
SMA_A19	
SMA_A20	
SMA_nADSC	
SMA_nADSP	
SMA_nADV	
SMA_nBW1	
SMA_nBW2	
SMA_nBW3	
SMA_nBW4	
SMA_nBWE	
SMA_nCE	
SMA_CLK	
SMA_nCS1	
SMA_nGW	
SMA_nLBO	
SMA_D31	
SMA_D30	
SMA_D29	
SMA_D28	
SMA_D27	
SMA_D26	
SMA_D25	
SMA_D24	
SMA_D23	
SMA_D22	
SMA_D21	
SMA_D20	
SMA_D19	
SMA_D18	
SMA_D17	
SMA_D16	
SMA_D15	
SMA_D14	
SMA_D13	
SMA_D12	
SMA_D11	
SMA_D10	
SMA_D9	
SMA_D8	
SMA_D7	
SMA_D6	
SMA_D5	
SMA_D4	
SMA_D3	
SMA_D2	
SMA_D1	
SMA_D0	

Y27	
AA28	
W29	
W30	
AA34	
AA35	
Y29	
AA30	
AC38	
AC39	
AA31	
Y31	
AC35	
AB35	
AB33	
AA33	
AC33	
AC34	
AD37	
AC37	
AC32	
AB31	
AE39	
AD39	
AE38	
AE39	
AD35	
AD36	
AC30	
AB30	
AE37	
AE38	
AB27	
AB28	
AE34	
AD34	
AD31	
AD32	
AE36	
AE36	
AJ39	
AH39	
AG37	
AG38	
IO_L22N_15	
AH37	
AH38	
AE34	
AE35	
AJ36	
AJ37	
AG35	
AG36	
AJ35	
AH35	
AK38	
AK39	
AM37	
AM38	
AL38	
AL39	
AM36	
AL36	
AK36	
AK37	

Bank 15

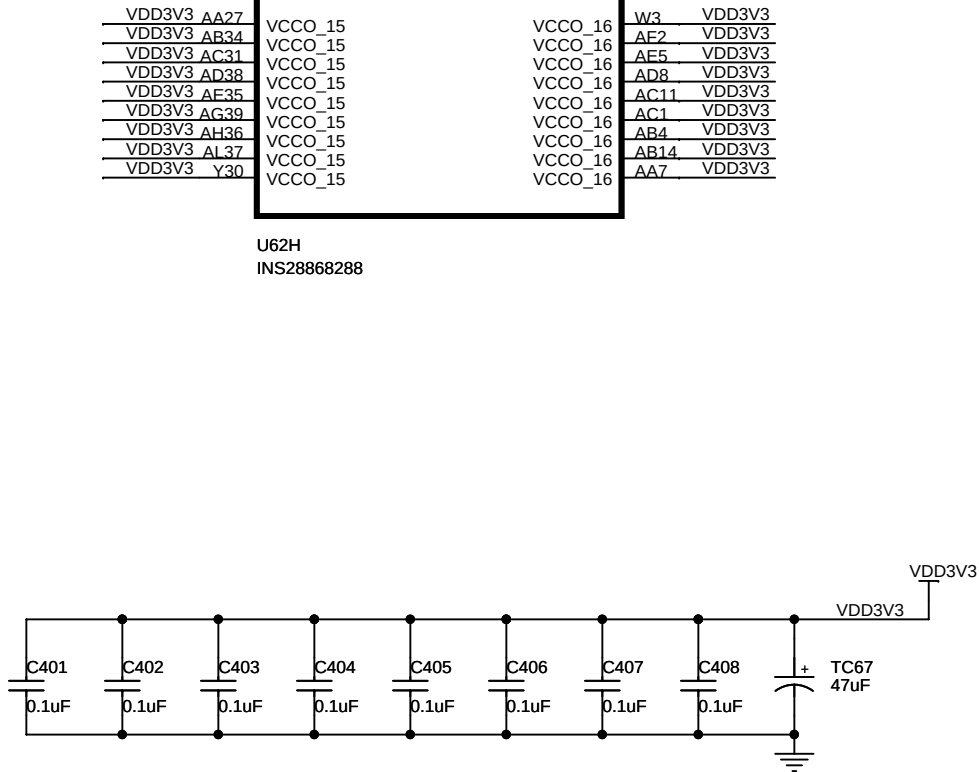
IO_L1P_15	
IO_L1N_15	
IO_L2P_15	
IO_L2N_15	
IO_L3P_15	
IO_L3N_15	
IO_L4P_15	
IO_L4N_VREF_15	
IO_L5P_15	
IO_L5N_15	
IO_L6P_15	
IO_L6N_15	
IO_L7P_15	
IO_L7N_15	
IO_L8P_CC_LC_15	
IO_L8N_CC_LC_15	
IO_L9P_CC_LC_15	
IO_L9N_CC_LC_15	
IO_L10P_15	
IO_L10N_15	
IO_L11P_15	
IO_L11N_15	
IO_L12P_15	
IO_L12N_VREF_15	
IO_L13P_15	
IO_L13N_15	
IO_L14P_15	
IO_L14N_15	
IO_L15P_15	
IO_L15N_15	
IO_L16P_15	
IO_L16N_15	
IO_L17P_15	
IO_L17N_15	
IO_L18P_15	
IO_L18N_15	
IO_L19P_15	
IO_L19N_15	
IO_L20P_15	
IO_L20N_VREF_15	
IO_L21P_15	
IO_L21N_15	
IO_L22P_15	
IO_L22N_15	
IO_L23P_VRN_15	
IO_L23N_VRP_15	
IO_L24P_CC_LC_15	
IO_L24N_CC_LC_15	
IO_L25P_CC_LC_15	
IO_L25N_CC_LC_15	
IO_L26P_15	
IO_L26N_15	
IO_L27P_15	
IO_L27N_15	
IO_L28P_15	
IO_L28N_VREF_15	
IO_L29P_15	
IO_L29N_15	
IO_L30P_15	
IO_L30N_15	
IO_L31P_15	
IO_L31N_15	
IO_L32P_15	
IO_L32N_15	

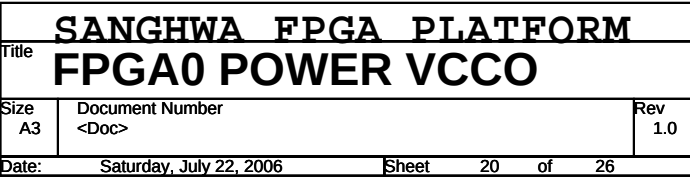
Bank 16

IO_L32N_16	
IO_L32P_16	
IO_L31N_16	
IO_L31P_16	
IO_L30N_16	
IO_L30P_16	
IO_L29N_16	
IO_L29P_16	
IO_L28N_VREF_16	
IO_L28P_16	
IO_L27N_16	
IO_L27P_16	
IO_L26N_16	
IO_L26P_16	
IO_L25N_CC_LC_16	
IO_L25P_CC_LC_16	
IO_L24N_CC_LC_16	
IO_L24P_CC_LC_16	
IO_L23N_VRP_16	
IO_L23P_VRN_16	
IO_L22N_16	
IO_L22P_16	
IO_L21N_16	
IO_L21P_16	
IO_L20N_VREF_16	
IO_L20P_16	
IO_L19N_16	
IO_L19P_16	
IO_L18N_16	
IO_L18P_16	
IO_L17N_16	
IO_L17P_16	
IO_L16N_16	
IO_L16P_16	
IO_L15N_16	
IO_L15P_16	
IO_L14N_16	
IO_L14P_16	
IO_L13N_16	
IO_L13P_16	
IO_L12N_VREF_16	
IO_L12P_16	
IO_L11N_16	
IO_L11P_16	
IO_L10N_16	
IO_L10P_16	
IO_L9N_CC_LC_16	
IO_L9P_CC_LC_16	
IO_L8N_CC_LC_16	
IO_L8P_CC_LC_16	
IO_L7N_16	
IO_L7P_16	
IO_L6N_16	
IO_L6P_16	
IO_L5N_16	
IO_L5P_16	
IO_L4N_VREF_16	
IO_L4P_16	
IO_L3N_16	
IO_L3P_16	
IO_L2N_16	
IO_L2P_16	
IO_L1N_16	
IO_L1P_16	

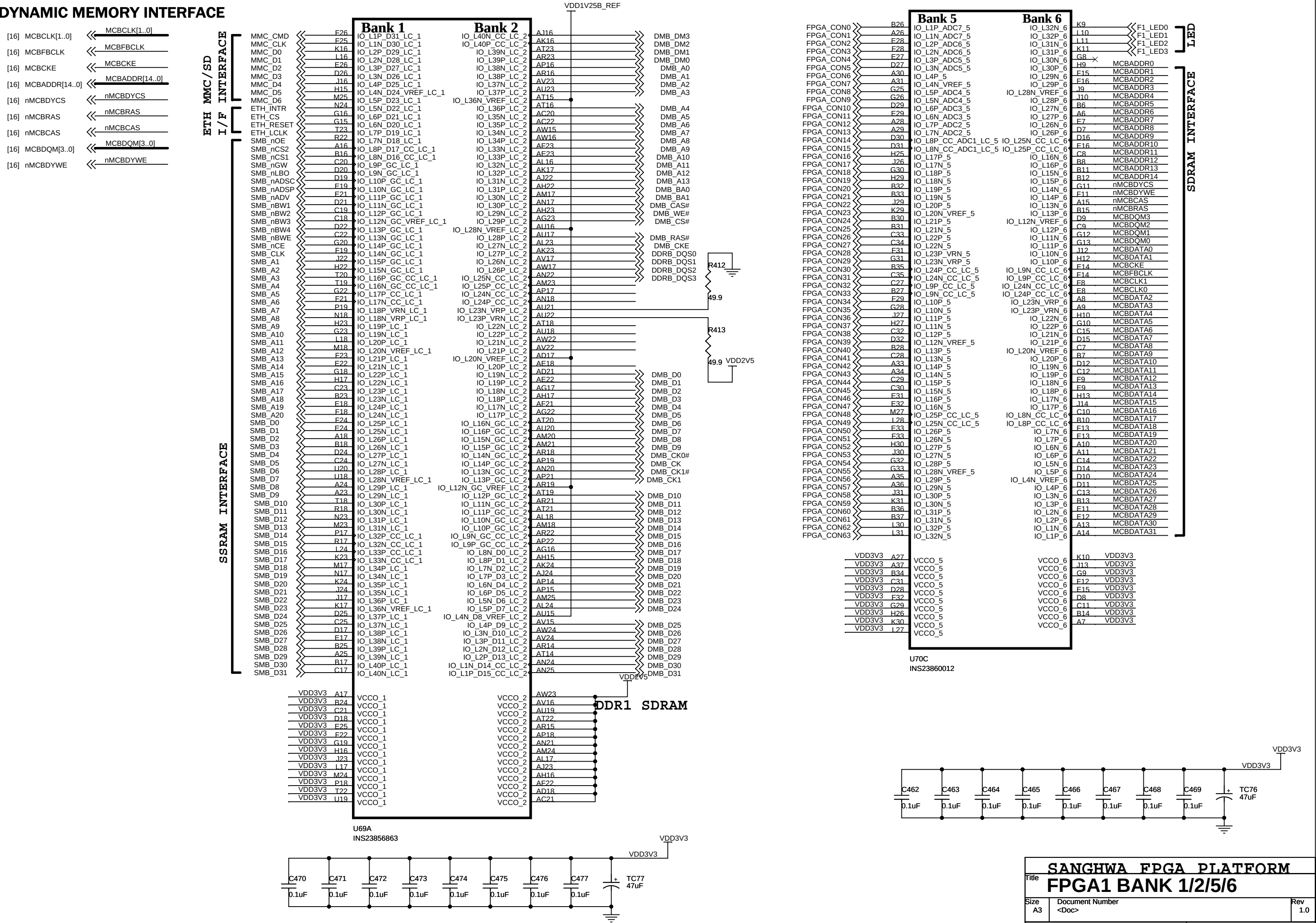
AF3	
AG3	
AE8	
AE9	
AG1	
AG2	
AE5	
AE6	
AD6	
AE6	
AE4	
AE4	
AE1	
AE1	
AD11	
AC12	
AE2	
AE3	
AC7	
AD7	
AD9	
AC10	
AC14	
AB15	
AC13	
AB13	
AD4	
AD5	
AD1	
AD2	
AB8	
AC8	
AC4	
AC5	
AB10	
AB11	
AA8	
AB7	
AC2	
AC3	
AB1	
AB2	
AB5	
AB6	
AA5	
AA6	
AA13	
AA14	
AA1	
Y1	
AB3	
AA3	
Y2	
Y3	
AA4	
Y4	
Y6	
Y7	
W1	
W2	
V2	
V3	
W4	
V4	

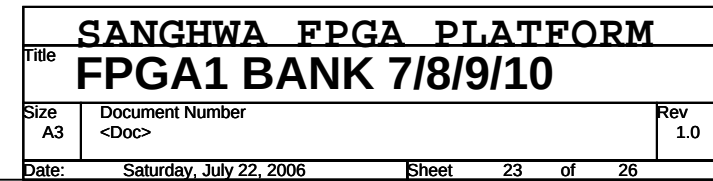
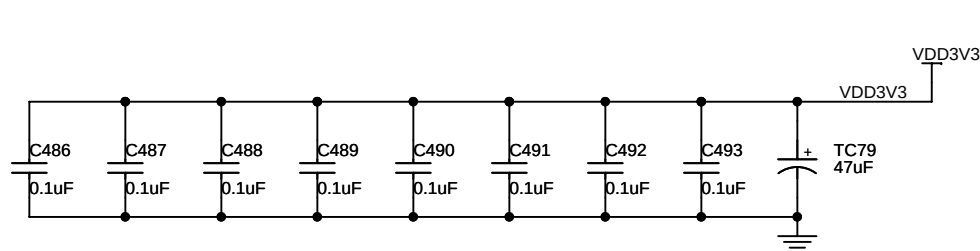
FPGA_CON192	
FPGA_CON193	
FPGA_CON194	
FPGA_CON195	
FPGA_CON196	
FPGA_CON197	
FPGA_CON198	
FPGA_CON199	
FPGA_CON200	
FPGA_CON201	
FPGA_CON202	
FPGA_CON203	
FPGA_CON204	
FPGA_CON205	
FPGA_CON206	
FPGA_CON207	
FPGA_CON208	
FPGA_CON209	
FPGA_CON210	
FPGA_CON211	
FPGA_CON212	
FPGA_CON213	
FPGA_CON214	
FPGA_CON215	
FPGA_CON216	
FPGA_CON217	
FPGA_CON218	
FPGA_CON219	
FPGA_CON220	
FPGA_CON221	
FPGA_CON222	
FPGA_CON223	
FPGA_CON224	
FPGA_CON225	
FPGA_CON226	
FPGA_CON227	
FPGA_CON228	
FPGA_CON229	
FPGA_CON230	
FPGA_CON231	
FPGA_CON232	
FPGA_CON233	
FPGA_CON234	
FPGA_CON235	
FPGA_CON236	
FPGA_CON237	
FPGA_CON238	
FPGA_CON239	
FPGA_CON240	
FPGA_CON241	
FPGA_CON242	
FPGA_CON243	
FPGA_CON244	
FPGA_CON245	
FPGA_CON246	
FPGA_CON247	
FPGA_CON248	
FPGA_CON249	
FPGA_CON250	
FPGA_CON251	
FPGA_CON252	
FPGA_CON253	
FPGA_CON254	
FPGA_CON255	

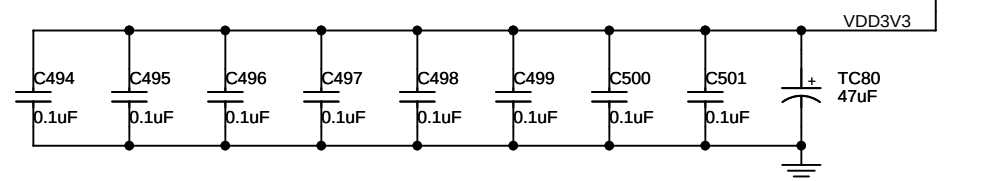
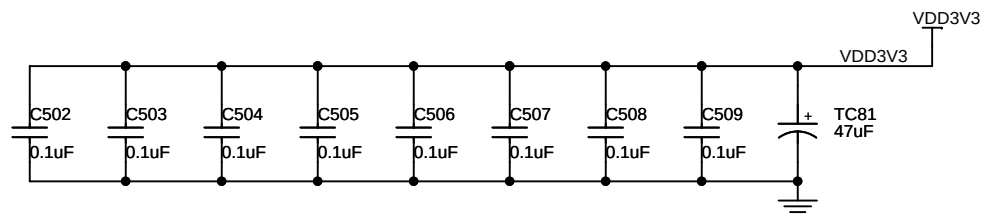
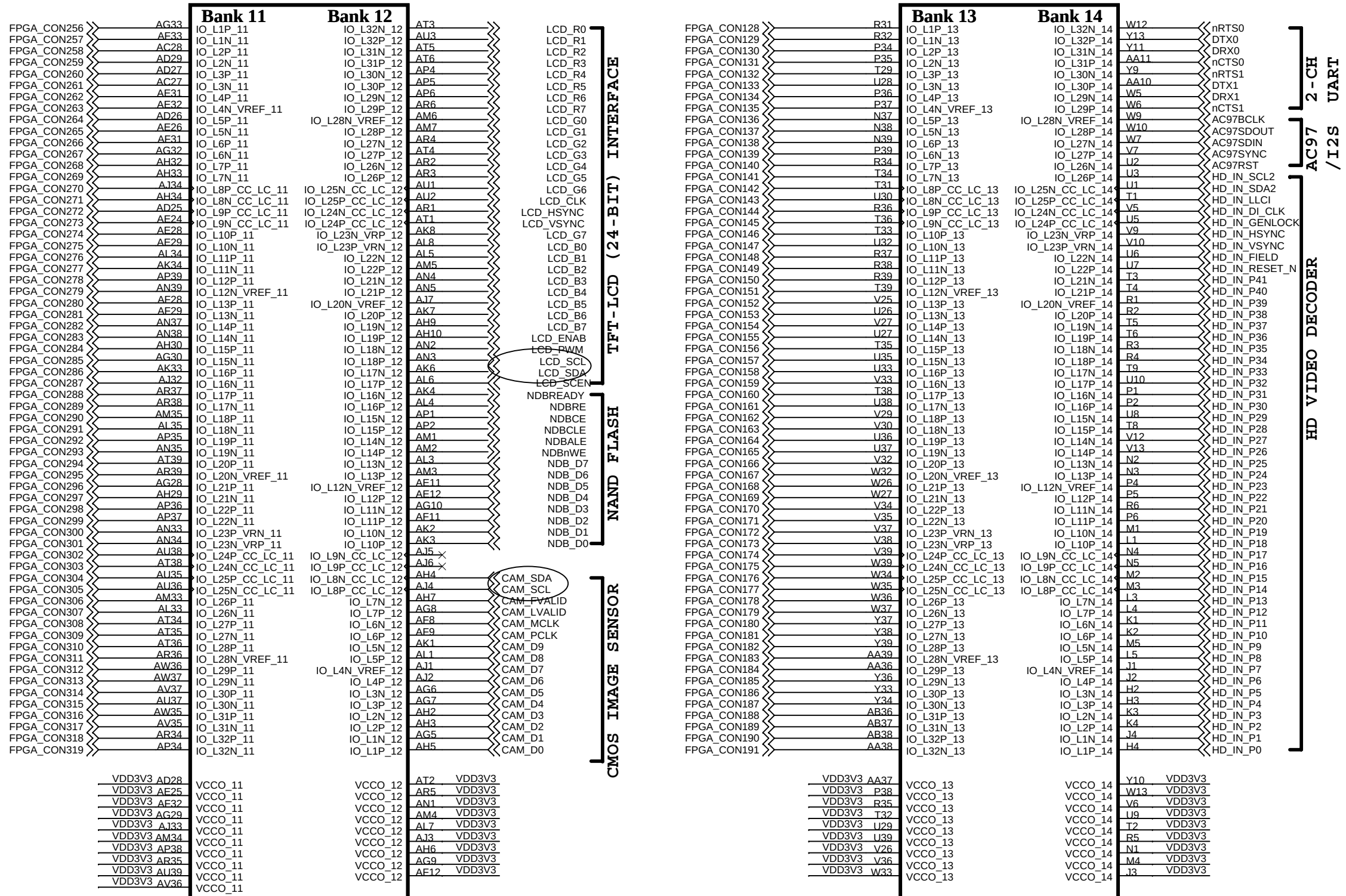




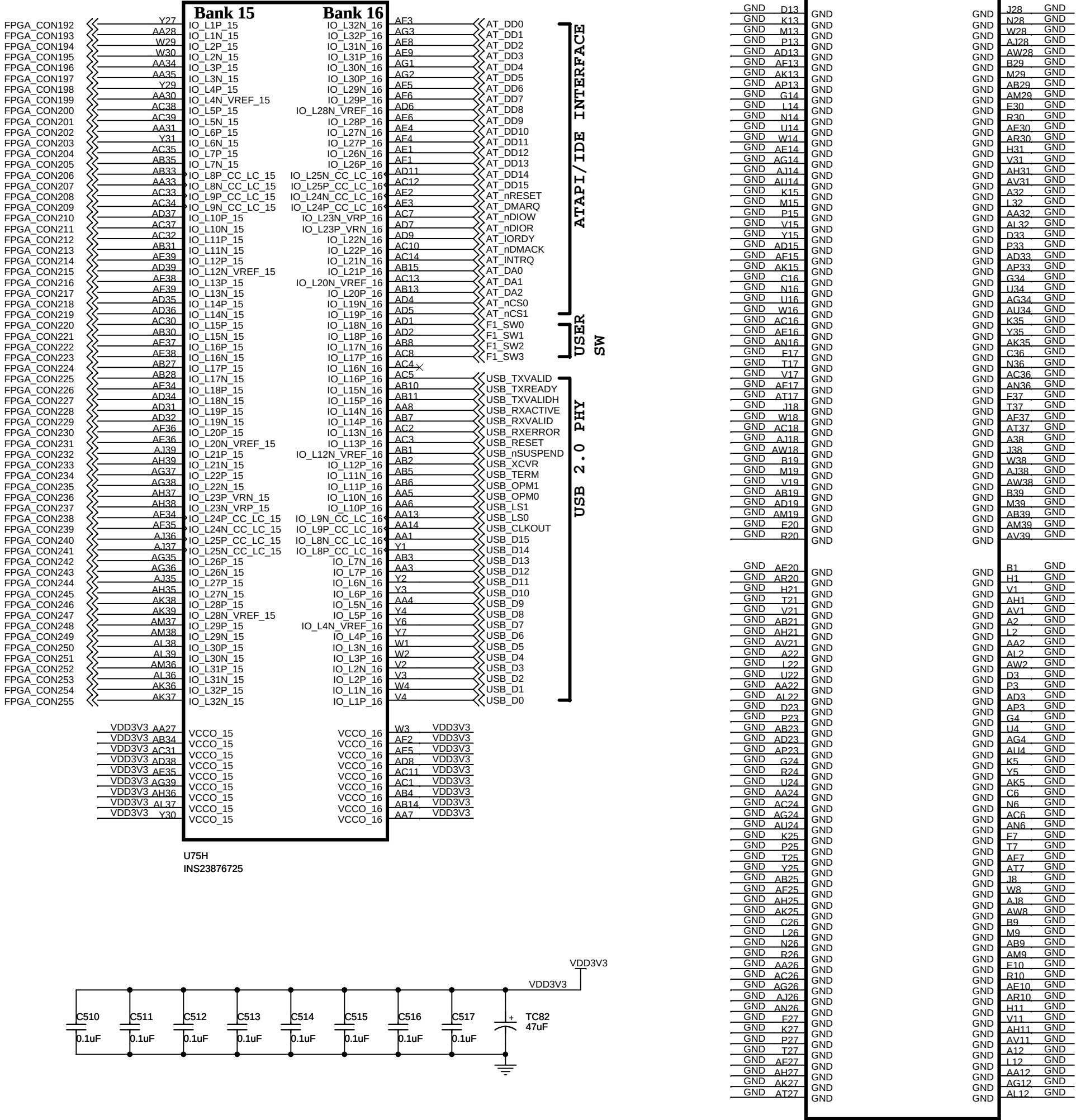
DYNAMIC MEMORY INTERFACE

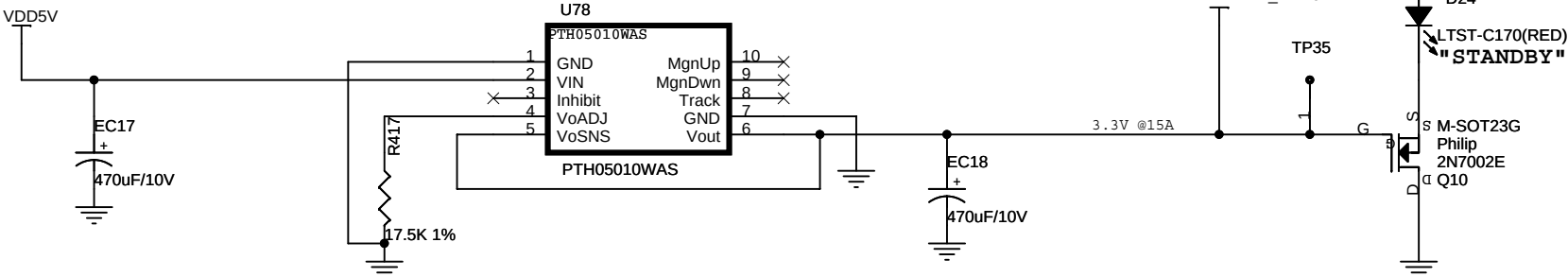






VIRTEX-4 FPGA
POWER GROUND





SANGHWA FPGA PLATFORM FPGA1 POWER VCCO			
Title	FPGA1 POWER VCCO		
Size A3	Document Number <Doc>	Rev 1.0	
Date: Saturday, July 22, 2006	Sheet 26	of 26	